WILL

Relational Geometry

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Abstract

This paper, the first in the **WILL RG Trilogy:**
Part I: Relational Geometry;
Part II: Relational Cosmology;
Part III: Relational Quantum Mechanics (R.Q.M.).
By applying extreme methodological constraints we establishing *Relational Geometry (RG)*: a background free foundation and its algebraic formulation **Relational Orbital Mechanics** that exactly reproduces the 1PN post-Newtonian expansions of General Relativity including perihelion precession, light deflection, and frame-dragging without the use of metric tensors, Christoffel symbols, or numerical geodesic integration, and produce specific $\mathcal{O}(\beta^4)$ correction that differs from 2PN GR and provides a falsifiable signature that we expect to be detectable in near future observations of the S2 star. From first principles we reproduce Lorentz factors, the energy-momentum relation, Minkowski and Schwarzschild metric intervals via the dimensionless projections of kinematic and potential state difference. All known GR critical surfaces (photon sphere, ISCO, horizons) emerge as simple fractions of this projections.

***This paper* offers new potential solutions to several long-standing problems, including:**

- **Derivation of Relational Orbital Mechanics** (Background-free and algebraically closed),
- **Resolution of GR singularities** (via bounded radial distance $r_{min} = R_s/2$ (4.9)),
- **Derivation of the equivalence principle** (from the rest-invariant state scaling (4.3, 4.2),
- **Removal of local energy density ambiguity** $\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$ (4.6.1),
- **Revelation of a clear relational symmetry between kinematic and potential** energy's (3.9),
- **Establishment of a computationally simpler and ontologically consistent foundation** (for subsequent papers on cosmology (Part II) and quantum mechanics (Part III)).

A Direct Note to the Reader: Why engage with this research?

Engaging with unknown independent research is a substantial commitment, and you should not do it blindly **I recommend the following:** Define the criteria this research must satisfy to justify the investment of your time. Then, ask WILL-AI if this work meets your standards. The AI is trained on this papers and will work as advance search engine. It will give you an honest, unbiased answer, within its limitations, and will provide direct links to related sections, colab notebooks and desmos projects. If you are still in doubt, browse **willrg.com**:

- **Falsifiable Predictions** - from atomic structure through local neighbourhood to cosmic horizon;
- **Galactic Dynamics** - test rotation curves predictions against **SPARC dataset**;
- **Holographic Decoder** - **Breaking** $M\sin(i)$ **Degeneracy** with 1D RV data string;
- **Computational Efficiency** - side by side comparison with General Relativity;
- **Logos Map** - trace the derivation logical chain;
- **WILL-AI** - just have a chat its fun.

Regardless of your decision, I wish you to find what you are seeking.

Anton Rize

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Following yet untamed thought

"There is no such thing as an empty space, i.e., a space without field. . . . Space-time does not claim existence on its own, but only as a structural quality of the field."
— *Albert Einstein, Relativity: The Special and the General Theory* (Appendix V: "Relativity and the Problem of Space"), 1952 edition, Methuen (London), p. 155; based on earlier 1920 additions.

IMPORTANT:

This document must be read **literally**. All terms are defined within the relational logic of WILL Relational Geometry.
Definition of Geometry: The term "geometry" is used herein in its fundamental, classical sense: the pure logic of relations, ratios, and structural constructions. It **does not** refer to modern differential geometry. There is no *a priori* metric tensor ($g_{\mu\nu}$), no Riemannian manifold, and no underlying coordinate grid. The geometry emerges from physical relations, not analytically from a background space.
Any attempt to reinterpret these derivations through conventional metric notions (*absolute distances, external backgrounds, hidden containers*) will produce distortions and mathematical misreadings.
Take the words as written and ontology as derived, without smuggling external concepts.

1 Stage 0: The Last Geocentric Epicycle

The standard formulation of General Relativity relies on the concept of a pre-existing, container-like spacetime that then gets "filled" with fields and matter.

Derivation of the Einstein field equations begins with the Einstein-Hilbert action, which is built upon the metric as the fundamental variable. This metric is defined on a smooth manifold that is assumed to exist *a priori*. This manifold, even without a metric, carries topological and differential structure - an absolute scaffold. This constitutes surplus ontology because it introduces an entity (the manifold) that is not derived neither observed.

The stress-energy tensor is derived from the variation of the matter Lagrangian with respect to the metric. This assumes that energy is a property of matter fields that can be localized in spacetime. However, this localization is frame-dependent (via the equivalence principle) and leads to well-known problems such as the non-uniqueness of the gravitational energy-momentum pseudotensor. Energy is treated as an absolute property of matter rather than a relational measure. This remains as metaphysical speculation in the heart of modern physics.

1.1 The contemporary split: an unjustified ontological commitment

All present-day theories (SR, GR, QFT, CDM, Standard Model) are built with a *bi-variable* syntax:

$$\underbrace{\text{fixed manifold} + \text{metric}}_{\text{structure}} \quad + \quad \underbrace{\text{fields} + \text{constants}}_{\text{dynamics}}.$$

No observation demands this duplication; it is retained purely because the resulting Lagrangians are empirically adequate *inside* the split. The split is therefore *not* an empirical discovery but an unjustified ontological statement.

1.2 Empirical deficit of the separation

- **Local energy conservation** verified only *after* the metric is declared fixed; no experiment varies the *volume* of flat space and checks calorimetry.
- **Universality of free fall** tests $m_i = m_g$ numerically, not the claim that inertia resides *in* the object rather than in a geometric scaling relation.
- **Gravitational-wave polarisations** test spin content, not ontology; extra modes can still be called "matter on spacetime".
- **Casimir/Lamb shifts** measure *differences* of vacuum energy between two geometries; the absolute bulk term is explicitly subtracted, leaving the split intact.

Every "test" is an *internal consistency check* of a formalism that already presupposes two substances. None constitute *positive evidence* for the split.

1.3 Historical Pattern: breakthroughs delete, not add

- **Copernicus** eliminated the Earth/cosmos separation.
- **Newton** eliminated the terrestrial/celestial law separation.
- **Maxwell** eliminated the electricity/magnetism separation.
- **Einstein** eliminated the space/time separation.

Each step widened the relational circle and reduced the number of primitives - an unexplained absolutes. The spacetime–energy split is the only survivor of this pruning sequence.

1.4 Consequence

Until an experiment varies the amount of space while holding everything else fixed, the spacetime–energy separation remains an *unevidenced metaphysical postulate.* **It is the last geocentric epicycle in physics.**

2 Stage I: Ontological Construction (The Primitives)

> *Every logical step is subject to empirical audit: if any consequence of this chain contradicts observation, the chain fails.*

As humans, we often project ourselves onto observed phenomena, obscuring the true nature of reality behind our anthropic tendencies. In order to prevent such misleading practices, we must establish extreme methodological constraints. Not as statements about the nature of reality, but strict rules of logical and epistemic purity. Without such methodology, any theoretical construct risks that excessive freedom will result in a framework that tells us more about ourselves than about the subject at study.

Primary Goal:

Derive the laws of physical reality from the most elementary principles by peeling-off interpretational layers until the foundational primitives revealed.

Traditionally a new theory is expected to propose new undetected fields or particles, postulates or axioms - a priori statements about physical reality. This research In contrast, adheres strictly to methodological constraints. The aim is for the theory to arise naturally from the geometry and logic itself. The foundational core of Relational Geometry - its principles and theorems - emerge from methodological constraints. They function as inescapable outcomes of the method itself.

The philosophical reasoning behind these constraints probably deserves its own dedicated publication. However, this research's main domain lies within physical reality that can be empirically tested; therefore, empirical alignment remains the ultimate test of the methods and the derived results they produce.

Flow chart of derivation logic

Logos Map

2.1 Foundational Methodological Principles

Principle 2.1 (Epistemic Hygiene)**.** Epistemic Hygiene as Refusal to Import Unjustified Assumptions. *This line of reasoning derive physics by* ***removing hidden assumptions****, rather than introducing new postulates. This construction is deliberate and contains zero free parameters. This is not a simplification - it is a deliberate epistemic constraint. No assumptions are introduced and no constructs are retained unless they are geometrically or energetically necessary.*

Principle 2.2 (Relational Origin)**.** All physical quantities must be defined by their relations. *Any introduction of absolute properties risks reintroducing metaphysical artefacts and contradicts the foundational insight of relationalism.*

Principle 2.3 (Ontological Minimalism)**.** *Maintain the minimum number of ontological assumptions and foundational primitives. Absolute background — spatial container, axis of time, absolute distance, forces, fields — are excluded as preexisting primitives but may appear as translation into common descriptive vocabulary.*

Principle 2.4 (Mathematical Transparency)**.**

1. *Every mathematical phrase, operational choice, or identity carries its own ontological statement.*
2. *Each mathematical object must correspond to explicitly identifiable relation between observers with transparent ontological origin.*
3. *Every symbol must be anchored to unique physical idea.*
4. *Introducing symbols without explicit necessity constitutes* ***Over-parameterization****: the proliferation of symbols without corresponding physical meaning.*
5. *Number of symbols = Number of independent physical ideas.*

   Mathematics is a language, not a world. Its symbols must never outnumber the physical meanings they encode.

A Note on Methodology vs. Ontology

It may be objected that the foundational principles constitute axioms, thereby contradicting the claim of *zero assumptions*. This objection conflates *methodological constraints* with *ontological axioms*.
Under the strictures of the Scientific Method, the burden of proof rests exclusively on the claimant asserting an absolute existence. Refusing to import unobserved absolutes (such as a background container or intrinsic properties) is not an assumption; it is the default epistemic stance — the null hypothesis.
These principles function strictly as rules of logical engagement to prevent the introduction of surplus ontology. To challenge them is to attempt to justify the importation of unsubstantiated metaphysical postulates, which contradicts the empirical foundation of physics.

2.1.1 The Foundational Core: Relational Geometry

This foundational methodological principles lead us to the core of Relational Geometry:

Theorem 2.5 (Relational Closure)**.** *A purely relational system cannot be geometrically open; Any stable, self-contained system must be described by a closed and self-consistent relational geometry. Its geometry is closed by absolute analytic necessity.*

Proof by Analytic Tautology.

1. **Premise 1 (Relational Origin):** Within RG Physical quantities are defined exclusively by their relations (Principle 2.2).
2. **Premise 2 (Interaction as Integration):** If a hypothetical "outside" entity or boundary were to physically interact with the system, that interaction itself constitutes a relation.
3. **Deduction (The Tautology):** As relation is formed, it is mathematically and physically integrated into the system.

An "open relational system" experiencing external influence is a logical contradiction. The system is closed not by a spatial boundary or a metric container, but by the strict impossibility of an interacting non-relation. The system is relationally closed by definition. □

2.1.2 What is Energy in Relational Ontology?

Across all domains of physics, one empirical fact persists: in every closed system there exists a quantity that never disappears or arises spontaneously, but only transforms in form. This invariant is observed under many guises — kinetic, potential, thermal, chemical — yet all are interchangeable, pointing to a single underlying structure.

Crucially, this quantity is never observed directly, but only through *differences between states*: a change of velocity, a shift in configuration, a transition of phase. Its value is relational, not absolute: it depends on the chosen frame or comparison, never on an object in isolation.

Moreover, this quantity provides continuity of causality. If it changes in one part of the system, a complementary change must occur elsewhere, ensuring the unbroken chain of transitions. Thus it is the bookkeeping of causality itself — the ledger of change.

Theorem 2.6 (Relational Invariance (Conservation))**.** *Within relationally closed system without an external background, the existence of change necessitates the existence of conserved invariant measure of change. Because the system is closed any change must be perfectly balanced by a complementary change elsewhere in the system conserving the total transformations.*

Proof by Structural Necessity.

1. **Premise 1 (Relational Origin):** Physical quantities are defined exclusively by their relations (Principle 2.2).
2. **Premise 2 (Empirical Change):** States undergo observable change.
3. **Premise 3 (Relational Closure):** As proven in Theorem 2.5, the relational system is analytically closed. There is no external background, void, or hidden container to absorb or inject influence.
4. **Deduction (Causal Continuity):** Because the system is closed (Premise 3), an empirical change (Premise 2) cannot vanish into a void or emerge from one. To maintain the structural integrity of the relations (Premise 1), any change must be perfectly balanced by a complementary change elsewhere in the system.

Therefore, within this closed relational network, there must exist an invariant quantitative measure of change. ☐

Corollary 2.7 (Historical Note on Energy). *In classical and relativistic mechanics, this derived invariant measure of change is assigned the historical label "Energy". By proving this invariance geometrically, we strip "Energy" of its substantialist properties (as a "substance" or "thing") and reveal its origin as the necessary measure of change and bookkeeping ledger of causality.*

Energy :

Definition 2.8 (Energy).

Energy is the invariant relational measure
of state transformation
within relationally closed system.

Remark 2.9 (Energy Ledger: intuitive understanding). *Consider a completely closed economy with no physical currency — no dollar bills, no gold coins, and no central bank (acting as an absolute container). It is a purely relational economy consisting solely of a massive ledger of debts and credits between individuals. If one account decreases, another must increase. If the system is frozen at any given moment, the sum of all debts and credits across the entire ledger must equal zero. "Money" does not exist as a physical object; it is simply the mathematical rule that the ledger must always balance. Energy functions identically: it is not a substance, but the invariant rule of relational accounting.*

Theorem 2.10 (Relational Isotropy). *Within a background-free relational system, no spatial direction or coordinate can be* a priori *privileged. Therefore, the geometry encoding the relational balance (energy) must be maximally symmetric (isotropic).*

Proof by Relational Origin.

1. **Premise 1 (Absence of a Grid):** As established by Theorem 2.5 (Relational Closure), the relational system possesses no external container, absolute metric tensor, or underlying coordinate grid.
2. **Premise 2 (Direction requires Relation):** Under the Principle of Relational Origin (Principle 2.2), "direction" or "orientation" cannot exist in a void; it only physically manifests as a specific interaction or measurement between defined participants.
3. **Deduction (Pure Gauge):** Because there is no absolute grid (Premise 1), assigning an intrinsic, absolute axis to the unmeasured relational ledger (energy) introduces unobservable "surplus ontology." Any such arbitrary orientation is a mathematical artifact (pure gauge).

Therefore, prior to a specific interaction defining a local axis, the pure relational structure encoding this conserved energy ledger must possess no intrinsic privileged direction. It must be strictly isotropic. ☐

2.2 Unifying Ontological Principle

Summary

Any attempt to treat *"spacetime structure"* as separate from *"dynamics"* smuggles in a background container that is not justified by the phenomena. This violates epistemic hygiene: it introduces an ontological artifact without necessity. Eliminating this separation compels the identification of structure and dynamics as two aspects of a single entity.

Lemma 2.11 (False Separation). *Any model that treats processes as unfolding* within *an independent background necessarily assigns to that background structural features (metric, orientation, or frame) not derivable from the relations among the processes themselves. Such a background constitutes an* extraneous absolute.

Proof. Suppose an independent background exists. Then at least one of its structural attributes - metric relations, a preferred orientation, or a class of inertial frames - remains fixed regardless of inter-process data. This attribute is not relationally inferred but posited a priori. It thereby violates the relational closure principle: it introduces a non-relational absolute external to the system. Hence the separation is illicit. ☐

Corollary 2.12 (Structure–Dynamics Coincidence). *Without introducing new primitives 2.11, we should see structure and dynamics as a single ontological element. The structural arena and the dynamical content must be identified: structure is dynamics, energy is spacetime.*

This leads us to principle with *negative ontological weight*: it reduces the number of primitives and removes the separation between structure and dynamics. Spacetime is not a "container" but relational tension between energy states.

Principle 2.13 (Unifying Ontological Principle)**.**

$$\boxed{\textbf{\textit{SPACETIME}} \equiv \textbf{\textit{ENERGY}}}$$

No background container permitted.

Definition 2.14 (WILL)**. WILL ≡ SPACE-TIME-ENERGY** *is the technical term we use for the unified relational structure determined by Unifying Principle 2.13. All physically meaningful quantities are relational features of WILL.*

The Last Geocentric Epicycle

Definition 2.15 (Ontological weight)**.** *The number of irreducible primitives a theory posits.*

Picture	Primitives
Substantival: container + content (spacetime, energy/matter)	≥ 2
Relational: WILL (one self-relating structure)	1

It is the operation physics has performed at every unification:

- **Heat** ≡ molecular motion (deletes caloric);
- **Electricity** ≡ magnetism (Maxwell);
- **Space** ≡ time (Einstein).

Each merged two primitives into one. **SPACETIME ≡ ENERGY** applies the identical operation. **This Principle does not add, it subtracts: it removes the hidden assumption. Structure and dynamics are two aspects of a single entity that we call WILL.**

Remark 2.16 (Auditability)**.** *Principle 2.13 is foundational but testable: it is subject to: (i) geometric audit (internal logical consequences) and (ii) empirical audit (agreement with empirical data).*

3 Stage II: Geometric Derivation

3.1 WILL Structure

Having established Principle 2.13 by removing the illicit separation of structure and dynamics, we established a single foundational primitive: ***WILL*** ≡ ***SPACE-TIME-ENERGY***. Now we proceed to derive the necessary geometric and physical properties of ***WILL***.

The foundational core (2.1.1) — Closure, Conservation, and Isotropy — establishes the necessary properties for the geometric carriers of the relational energy capacity.

Relational Carrier Conventions:

All references to "carriers" within WILL RG open research are to be read in the strict relational sense:

- **Degree of freedom (DOF):** An n-dimensional relational carrier encoding the conserved transformation capacity.
- **Direction:** An oriented relation. Opposite orientations are physically distinct and not identified.
- **Closed carrier:** Compact and without boundary; relational capacity cannot leak into an external void.
- **No background:** No external embedding space. All geometric structure is reconstructible solely from relations between participants.
- **Maximal symmetry:** The carrier is homogeneous and isotropic with respect to oriented directions.
- **Minimal relational carrier:** A closed, maximally symmetric carrier utilizing exactly the required DOF without arbitrary parameters.

3.2 Deriving Relational Carriers

Theorem 3.1 (Minimal Relational Carriers of the Conserved Relational Capacity). *The minimal relational carriers satisfying Closure, Conservation, Maximal Symmetry, and Mathematical Transparency are:*

(a) S^1 for directional (Kinematic) relational transformation;

(b) S^2 for omnidirectional (Potential) relational transformation.

Proof. The proof classifies the minimal types of relations and applies the derived geometric constraints:

(a) **1-DOF (Kinematic) Relation:** The minimal non-trivial relation involves 1 degree of freedom (1-DOF): a sequence of state transformations.

A linear segment cannot carry this relation because it possesses absolute endpoints. Endpoints act as boundaries where the transformational capacity would either vanish (violating the Conservation Theorem) or require an external bounding mechanism to reflect it (violating the Relational Origin and Ontological Minimalism principles).

To satisfy Theorem 2.5 (Closure) and Theorem 2.10 (Isotropy), the 1-DOF carrier must possess no boundaries and no privileged points. The unique 1-DOF boundaryless, isotropic topology is the closed loop (S^1).

(b) **2-DOF (Potential) Relation:** The next minimal relation involves 2 degrees of freedom (2-DOF): the omnidirectional distribution of the transformational capacity from a primary state.

By Theorem 2.5, this 2-DOF carrier must be boundaryless to prevent leakage into an external void. By Theorem 2.10, it must be maximally symmetric, ensuring an isotropic distribution of potential.

Topological classification of a boundaryless 2-carrier under the strict requirement of maximal symmetry eliminates all anisotropic surfaces (e.g., the torus T^2, which possesses distinct major and minor intrinsic axes). The unique boundaryless, maximally symmetric 2-DOF topology is the 2-sphere (S^2).

(c) **Higher-Dimensional Carriers (S^n for $n \geq 3$):** The mathematical existence of higher-dimensional maximally symmetric topologies does not justify their ontological inclusion. The observable categories of relation are exhausted by directional shift (captured by the minimal 1-DOF S^1) and omnidirectional distribution (captured by the minimal 2-DOF S^2). Positing an S^3 or higher-dimensional carrier requires the existence of an independent, unobserved third category of macroscopic relation. By Principle 2.4 (Mathematical Transparency), introducing geometric degrees of freedom without uniquely identifiable physical relations constitutes over-parameterization. Furthermore, because higher-dimensional spheres contain S^1 and S^2 as lower-dimensional sub-structures, adopting them as foundational primitives when the minimal topologies suffice strictly violates Principle 2.3 (Ontological Minimalism). Therefore, all $n \geq 3$ carriers are excluded.

Ontological Minimalism and Mathematical Transparency uniquely enforce S^1 and S^2 as the minimal relational carriers without invoking metric geometry *a priori*. □

WARNING: The Spatial Container Fallacy

S^1 and S^2 are strictly not to be interpreted as physical geometries embedded in a spacetime container.
Do not visualise a sphere or a ring expanding through a 3D void.

- S^2 has no physical surface area, no volume, and does not "dilute" a flux across a spatial distance.
- They are purely abstract, algebraic phase spaces — relational carriers encoding the closure, conservation, and isotropy of the transformational resource.

Spatial distance (r) and time are purely emergent descriptive labels that observers attach to these underlying algebraic phase patterns. The phase projection dictates the space; the space does not dictate the phase.

Remark 3.2 (Constructive Derivation Protocol). *This derivation proceeds by construction from traced origins. Each object entering the chain - the principles, the theorems, the carriers - has an explicit derivation path. Alternative structures (discrete groups, higher-dimensional carriers, non-orientable surfaces) are not excluded by enumeration; they are absent because no derivation chain within this framework produces them.*

3.3 The Amplitude-Phase Duality

Lemma 3.3 (Duality of Relation). *By the Principle of Relational Origin (2.2), the physical manifestation of a state cannot be quantified by a single absolute magnitude. The identification of spacetime with energy necessitates a complementary internal ratio of two measures:*

1. *the* ***amplitude*** *of transformation (external shift), and*
2. *the* ***phase*** *of transformation (internal order).*

Proof. By the Principle of Relational Origin (2.2), assigning a single unbounded scalar to describe a transformation introduces an unobservable absolute reference scale. A purely relational state must therefore be defined internally, as a ratio or partitioning of a closed resource.

The geometric carriers S^1 and S^2 mathematically enforce this requirement. Identifying a unique state on these closed carriers necessitates two orthogonal projections. A single projection cannot capture a ratio. One projection must quantify the extent of the external relation (Amplitude), which geometrically dictates an orthogonal complement quantifying the retained internal structure (Phase). Together, these dual, coupled axes provide the minimal, parameter-free ratio required to define a state transformation without invoking an external background. □

The orthogonal decomposition of the relational carriers S^1 and S^2 reveals a functional duality. Physical state is a combination of two projections:

Definition 3.4 (The Amplitude Projection (External Interaction))**.** *Denoted by β (kinematic) and κ (potential). This component measures the extent of external relation. It manifests as momentum or potential intensity. Physically, it represents the system's relational "shift" from the* ***Observer's Relational Origin*** *(the rest frame).*

$$\textit{Amplitude} \to \textit{External Power (Kinetic/Potential)}$$

Definition 3.5 (The Phase Projection (Internal Evolution))**.** *Denoted by β_Y and κ_X. This component measures the internal ordering. It governs the intrinsic scale of proper time and proper length. A Phase of 1 represents maximal internal flow (rest), while a Phase of 0 represents the collapse of internal causality (light-speed/event horizon).*

$$\textit{Phase} \to \textit{Internal Order (Time/Structure)}$$

Theorem 3.6 (Conservation of Relation)**.** *For both kinematic (S^1) and potential (S^2) modes, the sum of the squared Amplitude and squared Phase is invariant:*

$$\underbrace{\textit{Amplitude}^2}_{\textit{External Interaction}} + \underbrace{\textit{Phase}^2}_{\textit{Internal Existence}} = 1 \tag{1}$$

Proof. By Theorem 3.1, the balanced relational ledger is carried by the maximally symmetric geometries S^1 and S^2. Normalizing the total relational capacity to unity (1), any physical state corresponds to a point on these carriers. The algebraic identity of orthogonal projections on a unit circle and sphere dictates that the sum of their squares must equal the squared radius. Therefore, the parameters satisfy $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$, encoding the finite, closed relational ledger. □

The Projections are Cross-Cultural Invariants

- S^1 is the minimal geometric structure that can encode a directional relation without absolute space. Its parameter is an angle θ_1 whose cosine ($\beta = \cos(\theta_1)$) we identify with the fraction v/c and whose sine ($\beta_Y = \sin(\theta_1)$) governs the internal ordering (proper time fraction).
- S^2 is the minimal geometric structure that can encode omnidirectional relation without absolute space. Its parameter is an angle θ_2 whose sine ($\kappa = \sin(\theta_2)$) we identify with the escape velocity fraction v_e/c and whose cosine ($\kappa_X = \cos(\theta_2)$) governs the internal ordering (proper time fraction).

Any two observers assign β and κ the same value for the same state, whatever their unit system — or even their descriptive vocabulary. This mappings:

$$\beta = \frac{v}{c}, \qquad \kappa = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}} = \sqrt{\frac{\rho}{\rho_{max}}}$$

are *translations* into descriptive vocabulary.

3.3.1 Consequence: Relativistic Effects

Proposition 3.7 (Physical Interpretation: Relativistic Effects)**.** *The conservation law of Theorem 3.6 implies that any redistribution between the orthogonal components manifests physically as the relativistic effects of time dilation and length contraction.*

Proof. By Theorem 3.6, the components satisfy $\beta^2+\beta_Y^2=1$ and $\kappa^2+\kappa_X^2=1$. An increase in Amplitude (β,κ) enforces a geometric decrease in Phase measure (β_Y,κ_X). This necessary reduction of β_Y and κ_X corresponds precisely to the dilation of proper time and the contraction of proper length, while the growth of β and κ represents momentum or potential depth. Thus, the kinematic and potential trade-off is the direct, un-parameterized physical expression of the geometric closure of the S^1 and S^2 carriers. □

This identifies Relativistic and Gravitational Time Dilation not as a mysterious "slowing down" of clocks, but as geometric **phase rotation**. As a system invests more of its relational existence into external Amplitude (β or κ), it necessarily withdraws from its internal Phase (β_Y or κ_X), changing the rate of its proper time evolution.

Logical Chain Summary

↓*Methodological Constraints*↓
Relational: Closure + Conservation + Isotropy Theorems.
↓ *(apply to existing physics)*↓
Two Primitives: $\underbrace{\text{fixed manifold} + \text{metric}}_{\text{structure}} + \underbrace{\text{fields} + \text{constants}}_{\text{dynamics}}$
↓ *(Ontological reduction by Relational Origin Principle no background allowed)*↓
One Primitive: ***WILL*** $\equiv$ ***SPACE-TIME-ENERGY*** $\Longrightarrow$ **SPACETIME** $\equiv$ **ENERGY**
↓ *(by Closure + Conservation + Isotropy Theorems derive primal relational carriers)*↓
S^1 **(1-DOF kinematic)** + S^2 **(2-DOF potential)**
↓ *(By Duality of Relation Lemma)*↓
Relational Ledger: $\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1$
↓↓↓ $\underbrace{\beta^2+\beta_Y^2=1}_{\text{Kinematic}}$ on S^1 and $\underbrace{\kappa^2+\kappa_X^2=1}_{\text{Potential}}$ on S^2.

3.4 Relational Origin of Spatial Distance and the Inverse-Square Law

Spatial distance is derived as the operational inverse of relational phase capacity. Treating distance as an *a priori* foundational primitive violates the principles of Relational Origin (2.2) and Ontological Minimalism (2.3).

Theorem 3.8 (Inverse-Square)**.** *The inverse-square potential proportionality $\kappa^2 \propto 1/r$ is a geometric necessity of the S^2 carrier's relational capacity, determined without spatial primitives.*

Proof. Distance can not be *a priori* foundational primitive. It violates the principles of Relational Origin (2.2) and Ontological Minimalism (2.3). Therefore it must be defined by the relational tension between energy states. By the Spectroscopic Phase Shift Theorem, the potential projection κ^2 is operationally extracted entirely from the dimensionless redshift (z_κ), independent of spatial parameters.

Normalizing the absolute geometric scale at saturation ($\kappa^2 = 1 \equiv r = R_s$), spatial distance r emerges as the reciprocal measure of the omnidirectional projection amplitude on S^2:

$$\kappa = \sin(\theta_2) \Longrightarrow \kappa^2 = \frac{R_s}{r} \Longrightarrow r = \frac{R_s}{\kappa^2} \Longrightarrow \kappa^2 \propto 1/r. \tag{2}$$

This is a direct realization of the Unifying Principle (2.13). The $1/r$ proportionality is an analytic identity defining spatial distance, not an empirical input. □

Remark 3.9. *Here and in all RG papers the causal horizon $\kappa = 1$ is symbolised with usual letters for Schwarzschild radius R_s for translational ease. Despite there ontological difference this two horizons are mathematically identical. In order to keep the number of introduced new symbols at minimum the historical symbols remain. The complete independence of causal horizon $\kappa = 1$ from its historical origin $R_s = \frac{2GM}{c^2}$ is proven in R.O.M. sections sec:spectroscopic_shift, sec:absolute_scale, sec:relational_parameterization, sec:Kerr and others.*

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Summary

The inverse-square proportionality does not exist because physical space dictates how forces dilute. It exists because "spatial distance" is the descriptive vocabulary assigned to the inverse capacity of the omnidirectional relational carrier S^2.

The geometry of spacetime is dictated by the Relations of energy states.

3.5 Kinetic Energy Projection β on S^1

- **The Amplitude Component ($\beta = v/c$):** External shift or the extent of kinematic relation observed from an arbitrary relational origin (the observer's rest frame). It quantifies how much of the universal "transformation rate" is perceived as motion through space, relative to the observer. In legacy units, this is identified as the dimensionless ratio of velocity to the speed of light.
- **The Phase Component ($\beta_Y = \sqrt{1-\beta^2}$):** Internal ordering or the intrinsic scale of proper time and proper length within the moving system. A phase of 1 signifies maximal internal flow (rest), while a phase of 0 indicates the collapse of internal causality (light-speed).

These two components are bound by the conservation theorem (3.6) in closed system, which acts as a strict "ledger of transformation":

$$\beta^2 + \beta_Y^2 = 1$$

The manifestation of any system is distributed between its internal (Phase) and relational (Amplitude) aspects.

Since S^1 encodes one-dimensional shift, the total energy E of the system must project consistently onto both external and internal axes:

$$E^2 = E_X^2 + E_Y^2$$

where

$$E_X = E\beta, \qquad E_Y = E\beta_Y.$$

Desmos project

Kinetic projection β on the S^1 carrier

3.5.1 Geometric Nature of Rest Energy State

Theorem 3.10 (Invariant Projection of Rest Energy). *For any state (β, β_Y) on the relational circle S^1, the total energy E must scale such that its internal projection remains constant and equal to the rest energy state E_0.*

$$E\beta_Y = E_0.$$

Proof. By the Relational Origin Principle (2.2), the internal structure of system A is determent by its rest frame $(\beta = 0, \beta_Y = 1)$ as the rest energy E_0. When an external frame B observes system A in relative motion $(\beta > 0)$, the geometric closure of the S^1 carrier necessitates a phase rotation $(\beta_Y < 1)$. If the total observed energy E remained static $(E = E_0)$, the projected internal energy $E_Y = E\beta_Y$ would decrease, implying that relative observation physically diminishes the internal causality of the observed system. This violates the structural integrity of the relation. To maintain the invariant internal reality of system A across all reference frames, the internal projection measured by B must perfectly equate to the rest state:

$$E_Y = E\beta_Y = E_0$$

Geometric consistency therefore dictates that the total energy E scales to compensate for the phase parameter:

$$E = \frac{E_0}{\beta_Y}$$

□

Summary:

The historical Lorentz factor γ is the reciprocal of β_Y. $\gamma = 1/\beta_Y$

3.5.2 The Geometric and Historical Nature of Mass

Corollary 3.11 (Rest Energy and Mass Equivalence)**.** *This geometry does not require mass as an independent concept. The invariant rest energy E_0 is numerically identical to mass m in natural units ($c = 1$), and is simply expressed in anthropocentric units (kilograms) via the conventional identity:*

$$E_0 = mc^2. \tag{3}$$

Proof. From the invariant projection $E\beta_Y = E_0$ and closure of S^1, no additional scaling parameter is required. Hence the conventional bookkeeping identities $E_0 = mc^2$ or $m = E_0/c^2$ reduce to tautologies. Mass is therefore not independent, but the rest-energy invariant itself. □

> Summary:
>
> **Mass is the invariant projection of total rest energy expressed in anthropocentric units of kilograms.**

3.5.3 Energy–Momentum Relation

Proposition 3.12 (Horizontal Projection as Momentum)**.** *On the relational circle, the unique linear geometric projection measure of external amplitude from rest is the horizontal projection $E_X = E\beta$; hence*

$$p \equiv E_X \equiv E\beta \quad (c = 1).$$

Proof. The rest state is $(\beta, \beta_Y) = (0, 1)$. A shift measure must (i) vanish at rest, (ii) grow monotonically with $|\beta|$, and (iii) flip sign under $\beta \mapsto -\beta$. The only relational candidate satisfying (i)-(iii) is the horizontal projection $E_X = E\beta$. Thus the identification is necessary and unique within methodological constraints. □

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>
> **Energy-momentum Triangle**

Corollary 3.13 (Energy–Momentum Relation)**.** *With p identified by Proposition 3.12 and $m = E_0$, the closure identity yields*

$$E^2 = p^2 + m^2 \quad (c = 1).$$

Equivalently, upon restoring c,

$$E^2 = (pc)^2 + (mc^2)^2.$$

Proof. By Conservation Theorem (3.6), $(E\beta)^2 + (E\beta_Y)^2 = E^2$. Substituting $p = E\beta$ and $m = E_0$ proves the claim. Restoring c is dimensional bookkeeping: $p \mapsto pc$ and $m \mapsto mc^2$, while E remains E, yielding the standard form. □

Remark 3.14 (Geometric Forms)**.** *The same identity may be expressed explicitly in terms of circle coordinates:*

$$E^2 = (\tfrac{\beta}{\beta_Y} E_0)^2 + E_0^2 = \big(\cot(\theta_1)\, E_0\big)^2 + E_0^2.$$

These are equivalent renderings of the same geometric necessity.

Remark 3.15 (Units sanity check - bookkeeping)**.** *In natural units ($c = 1$), the identification is $p \equiv E\beta$. Restoring c via dimensional bookkeeping ($p \mapsto pc$) yields:*

$$pc = E\beta = \frac{E}{c} v \implies p = \frac{E}{c^2} v. \tag{4}$$

With $E = \gamma mc^2$, this reduces to $p = \gamma mv$, the standard relativistic momentum. No new parameters are introduced.

$\beta = v/c \quad \theta_1 = \arccos(\beta)$	
Algebraic Form	**Trigonometric Form**
$\beta = v/c = \sqrt{1 - {\beta_Y}^2}$	$\beta = \cos(\theta_1)$
$\beta_Y = \sqrt{1 - (v/c)^2} = \sqrt{1 - \beta^2}$	$\beta_Y = \sin(\theta_1)$

Table 1: Geometric representation of relativistic effects.

> Summary
>
> **The energy–momentum relation $E^2 = (pc)^2 + (mc^2)^2$ is a geometric identity of S^1 carrier.**

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3.6 Minkowski Interval as Inflation of the S^1 Closure

In legacy Special Relativity the Minkowski interval is posited as the physical metric of a 4D background manifold. We reverse the order of explanation. The primitive object is the dimensionless kinematic closure on the relational carrier S^1 (Theorem 3.6):

$$\beta^2 + \beta_Y^2 = 1. \tag{5}$$

This is a single, parameter-free relation between the external (amplitude) and internal (phase) projections of the conserved transformation ledger. It invokes no coordinates, no container, and no external time. The Minkowski interval is *not* more fundamental than (5); it is what (5) becomes once a sequence of substantival posits is **added**, none of which is required by the relations themselves.

The inflationary posits. Beginning from the closure (5), legacy formalism introduces, in order:

(i) **A container.** The relation is declared to unfold *within* a pre-existing spatial manifold.

(ii) **Spatial coordinates.** Orthogonal labels x, y, z are imposed on the container to localize the amplitude projection.

(iii) **Externally flowing time.** An independent coordinate time t is posited and granted an autonomous flow, demoting the internal phase β_Y to a mere ratio of proper to coordinate time, $\beta_Y \equiv d\tau/dt$.

(iv) **A coordinate reference scale.** The combination c^2dt^2 is adopted as the magnitude against which the relational balance is measured.

The inflation. Under these posits the projections take their coordinate forms. The amplitude becomes the coordinate speed fraction distributed over the three container axes,

$$\beta^2 = \frac{v^2}{c^2} = \frac{dx^2 + dy^2 + dz^2}{c^2\, dt^2}, \tag{6}$$

and the phase becomes the time ratio $\beta_Y = d\tau/dt$. Substituting both into the closure (5),

$$\frac{dx^2 + dy^2 + dz^2}{c^2\, dt^2} + \left(\frac{d\tau}{dt}\right)^2 = 1, \tag{7}$$

multiplying both sides by c^2dt^2:

$$\frac{dx^2 + dy^2 + dz^2}{c^2dt^2} + \left(\frac{d\tau}{dt}\right)^2 = 1 \quad \xrightarrow{\times c^2dt^2} \quad dx^2 + dy^2 + dz^2 + c^2d\tau^2 = c^2dt^2$$

and clearing the posited denominator c^2dt^2 returns

$$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} \quad \Longrightarrow \quad \underbrace{c^2d\tau^2 = c^2dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}. \tag{8}$$

This is the Minkowski interval exactly. The legacy translation mapping is therefore:

- **Kinematic amplitude:** $\beta \equiv \cos\theta_1 = \dfrac{v}{c}$;
- **Internal phase:** $\beta_Y \equiv \sin\theta_1 = \dfrac{d\tau}{dt} = \dfrac{1}{\gamma}$.

Epistemological Consequence: the Interval is the Inflated Form

The Minkowski interval is recovered without remainder — but only after four substantival structures (a container, three spatial coordinates, an autonomously flowing time, and a coordinate reference scale) have been appended to $\beta^2 + \beta_Y^2 = 1$. Not one of these is demanded by the relations; each is surplus ontology. The Lorentz factor $\gamma = 1/\beta_Y = 1/\sin\theta_1$ and kinematic time dilation ($\beta_Y < 1$) are accordingly not deformations of a spatial fabric but a phase rotation on the unit circle S^1. The interval carries more structure than the relation it encodes; the relation, not the interval, is the irreducible fundamental primitive.

3.7 Potential Energy Projection κ on S^2

IMPORTANT:

Throughout this work, S^1 and S^2 are not to be interpreted as spacetime geometries but purely as relational carriers encoding energy conservation. Any reading otherwise is a misinterpretation.

Analogous to S^1, the relational geometry of the sphere S^2 provides orthogonal projections for two aspects of omnidirectional transformation. We define them as follows:

- **The Amplitude Component ($\kappa = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}}$):** The *relational potential measure* between the object and the observer. It corresponds to the *extent* of transformation as potential depth. Same way as we did with β on S^1, we will map κ on S^2 as fraction of speed of light. A value of $\kappa = 1$ denotes *saturation*: the entire relational resource of the system has been allocated into the potential channel. This condition mathematically defines the relational horizon where the escape velocity is equal to speed of light ($v_e = c$) or equivalently as radial distance equal to Schwarzschild radius ($r = R_s$ event horizon).
- **The Phase Component (κ_X):** This projection governs the intrinsic scale of proper length and proper time units within the potential field, corresponding to the internal *sequence* of transformation.

These two components are bound by the conservation theorem (3.6) in closed system, which acts as a finite "budget of transformation":

$$\kappa_X^2 + \kappa^2 = 1$$

The manifestation of any system is distributed between its internal (Phase) and relational (Amplitude) aspects. This single geometric constraint gives rise to the core gravitational phenomena of modern physics.

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Potential projection as κ on S^2 carrier

3.7.1 Potential Meridional Section of S^2

By isotropy, the omnidirectional carrier is S^2, but any radially symmetric exchange reduces to a great-circle meridional section. We therefore work on a unit great circle of S^2 with the parameterization $(\kappa_X, \kappa) = (\cos\theta_2, \sin\theta_2)$.

3.7.2 Consequence: Gravitational Effects

The redistribution of the relational ledger between the Phase and Amplitude components directly produces the effects of General Relativity. An increase in the relational amplitude measure (κ, gravitational potential) strictly necessitates a geometric decrease in the measure of the internal structure (κ_X). This geometric trade-off is observed physically as gravitational length contraction and time dilation. Thus, the curved geometry of spacetime is merely the shadow cast by the geometry of conserved relations.

3.7.3 Gravitational Tangent Formulation

Just as the relativistic energy–momentum relation can be expressed in terms of the kinematic projection $\beta = v/c$, we may construct its gravitational analogue using the potential projection $\kappa = v_e/c$, where v_e is the Newtonian escape velocity at radius r.

In the kinematic case, with $\beta = \cos\theta_1$, the energy relation can be written as

$$E^2 = \left(\cot\theta_1\, E_0\right)^2 + E_0^2, \tag{9}$$

so that the relativistic momentum is expressed as

$$p = (E_0/c)\cot\theta_1. \tag{10}$$

In full geometric symmetry, the gravitational case follows from $\kappa = \sin\theta_2$. We define the scalable gravitational total energy as

$$E_\kappa = \frac{E_0}{\kappa_X}, \qquad \kappa_X = \sqrt{1-\kappa^2}, \tag{11}$$

and introduce the gravitational analogue of momentum (potential momentum):

$$p_\kappa \;=\; (E_0/c)\tan\theta_2. \tag{12}$$

This yields the exact gravitational energy–momentum relation:

$$E_\kappa^2 = (p_\kappa c)^2 + (mc^2)^2. \tag{13}$$

Remark 3.16 (Ontological Status of Gravitational Momentum (p_κ)). *The gravitational momentum p_κ does not constitute a new empirical observable distinct from standard kinematic measurements; rather, it is the* ***ontological reinterpretation*** *of gravitational energy-momentum within the relational ontology.*

- ***Empirical Grounding:*** *p_κ is defined by the operationally measurable gravitational redshift z_κ, exactly as kinematic momentum p is defined by the transverse Doppler shift z_β (R.O.M. Spectral Shifts Derivation).*
- ***Geometric Equivalence Principle:*** *For a system in radial free-fall, the kinematic and potential amplitudes equalize ($\beta = \kappa$), yielding $p = p_\kappa$. This numerical identity is not a postulated equivalence, but a geometric necessity derived from the parallel closures of S^1 and S^2 and confirmed by Unified Relational Scaling Lemma (4.1).*
- ***Resolution of the Pseudotensor Problem:*** *Standard GR fails to localize gravitational energy because it treats energy as a property of a spatial region (the metric field). Because p_κ is relational measure between physical observers (quantified by spectroscopic phase shift), it requires no spatial container. It replaces the unlocalizable gravitational pseudotensor of legacy physics with a clean, operationally defined relational ledger.*

Summary:

$$\beta = \cos\theta_1, \qquad \kappa = \sin\theta_2,$$
$$\beta \;\longleftrightarrow\; \kappa, \qquad \cot\theta_1 \;\longleftrightarrow\; \tan\theta_2.$$

Kinematic momentum p and gravitational momentum p_κ are thus dual projections of the closed relational geometry, expressed through complementary trigonometric forms.

3.8 Schwarzschild Interval as Inflation of the S^2 Closure

The gravitational construction is strictly parallel to kinetic. The primitive object is the dimensionless potential closure on the relational carrier S^2 (Theorem 3.6):

$$\kappa^2 + \kappa_X^2 = 1, \tag{14}$$

relating the external potential amplitude κ to the internal phase κ_X. As before, (14) contains no radial coordinate, no central body, and no external time. The Schwarzschild component g_{tt} is recovered only by appending substantival posits.

The inflationary posits.

(i) **A container.** A spatial manifold is posited around a central body.

(ii) **A static observer.** The observer is held fixed in this container ($dr = d\theta = d\phi = 0$), so that only the temporal relation survives.

(iii) **Externally flowing time.** An independent coordinate time t is posited with autonomous flow, demoting the internal phase to the ratio $\kappa_X \equiv d\tau/dt$.

(iv) **A radial coordinate and legacy constants.** The dimensionless amplitude is localized as a coordinate function, $\kappa^2 \equiv R_s/r$, thereby introducing a radial coordinate r together with the legacy constants G and M through $R_s = 2GM/c^2$ — none of which is intrinsic to κ.

Note on Epistemic Hygiene. Posit (iv) is itself a legacy construction: κ is the primitive, while r, G, and M are coordinate-and-unit artifacts attached to it. In WILL R.O.M. (sections on relational parameterization, operational scale, and G) gravitational systems are described fully *without* G and *without* M, confirming that the projection κ, not R_s/r, is the irreducible primitive.

The inflation. Substituting $\kappa_X = d\tau/dt$ and $\kappa^2 = R_s/r$ into the closure (14),

$$\left(\frac{d\tau}{dt}\right)^2 + \frac{R_s}{r} = 1 \implies \left(\frac{d\tau}{dt}\right)^2 = 1 - \frac{R_s}{r}, \tag{15}$$

and introducing the posited reference scale $c^2 dt^2$ gives

$$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} \implies \underbrace{c^2 d\tau^2 = c^2 \left(1 - \frac{R_s}{r}\right) dt^2,}_{\text{mathematical scaffolding}} \tag{16}$$

which is the Schwarzschild interval for a static observer ($ds^2 = c^2 d\tau^2$). The legacy translation mapping is:

- **Potential amplitude:** $\kappa \equiv \sin\theta_2 = \sqrt{\frac{R_s}{r}}$;
- **Internal phase:** $\kappa_X \equiv \cos\theta_2 = \sqrt{1-\kappa^2} = \frac{d\tau}{dt}$.

Remark 3.17 (Full derivation)**.** *This covers the static interval. Full derivation of dynamic Schwarzschild and Kerr metrics we present within addition: R.O.M. (sec:Schwarzschild) , R.O.M. (sec:Kerr) .*

Two Inflation's

The Minkowski and Schwarzschild intervals are produced by the identical procedure: append a container, coordinates, an autonomously flowing time, and a coordinate reference scale to a single dimensionless closure — $\beta^2 + \beta_Y^2 = 1$ on the kinematic carrier, $\kappa^2 + \kappa_X^2 = 1$ on the potential carrier. Gravitational time dilation ($\kappa_X < 1$) is, like its kinematic counterpart, a phase rotation, not the curvature of a pre-existing fabric. What General Relativity treats as the fundamental metric of spacetime is the coordinate-laden image of a parameter-free relation. The relation is primary; the metric is its inflation.

3.9 Clear Relational Symmetry Between Kinematic and Potential Projections

On the unit kinematic circle (S^1) we parameterize

$$(\beta, \beta_Y) = (\cos\theta_1,\ \sin\theta_1),$$

so that the invariant internal phase projection reads

$$E\,\beta_Y = E_0 \implies \boxed{E = \frac{E_0}{\beta_Y} = \frac{E_0}{\sin\theta_1}}, \qquad p = \frac{E}{c}\,\beta = \frac{E_0\,\beta}{\beta_Y} = E_0 \cot\theta_1,$$

and therefore $E^2 = (pc)^2 + E_0^2$.

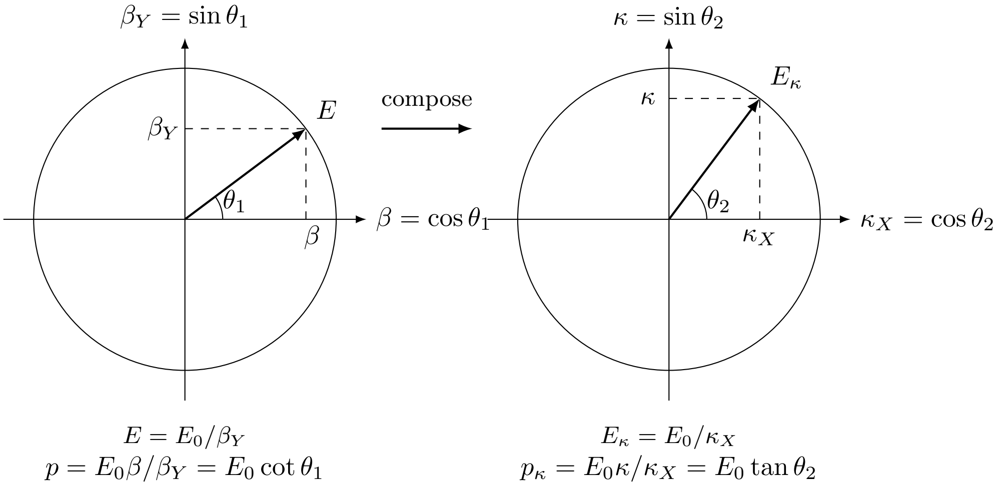
On the gravitational circle (S^2) we parameterize

$$(\kappa_X, \kappa) = (\cos\theta_2,\ \sin\theta_2),$$

so that the invariant internal phase projection reads

$$E_\kappa\,\kappa_X = E_0 \implies \boxed{E_\kappa = \frac{E_0}{\kappa_X} = \frac{E_0}{\cos\theta_2}}, \qquad p_\kappa = E_\kappa\,\kappa = \frac{E_0\,\kappa}{\kappa_X} = E_0 \tan\theta_2,$$

and therefore $E_\kappa^2 = p_\kappa^2 + E_0^2$.



Now we can clearly see the underlying symmetry between relativistic and gravitational factors that can be expressed in unified algebraic and trigonometric forms, as shown in Table 2.

3.10 Closure: Relational Origin of the Virial Theorem

Having established that directional (kinematic) and omnidirectional (potential) relations are carried by the S^1 and S^2 respectively, we now derive the relationship that unifies them.

3.10.1 Derivation of the Energetic Exchange Rate

Remark 3.18 (From slice to whole S^2). *Although we parametrise a single meridional great circle (κ_X, κ) for algebraic convenience, the amplitude κ^2 denotes the* total active omnidirectional ledger *over the S^2 carrier. The exchange-rate factor 2 reflects that S^2 has two independent relational degrees of freedom even when calculations are carried out on a representative great-circle section.*

3.10.2 Uniqueness of the Exchange Rate (No Hidden Weighting)

Lemma 3.19 (DOF-Indifference). *By maximal symmetry (no privileged directions, Theorem 2.10) and ontological minimalism (no hidden structure, Principle 2.3), each independent degree of freedom (DOF) has equal relational weight. WILL has no preferred directions and no preferred degrees of freedom.*

Proof. Any unequal weights to DOF's would create a preferred direction on maximally symmetric carriers violating the **Relational Isotropy Theorem** (2.10). It would require an extra rule that distinguishes the carriers on grounds other than their internal dimensions. Such a rule would constitute hidden structure, violating all four methodological principles (2.2, 2.3, 2.4, 2.1). Therefore, each independent DOF must receive an identical relational weight l. □

Theorem 3.20 (Closure). *In a relationally closed system, the kinematic carrier S^1 (1 DOF) and the potential carrier S^2 (2 DOF) exchange the active relational ledger in proportion to their degrees of freedom. Consequently, any relationally closed system must globally satisfy*

$$\boxed{\kappa^2 = 2\beta^2}$$

Proof. By Conservation Theorem (3.6) external amplitudes (β, κ) only in there quadratic forms complete there ledger ($\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1$). Let l be the active relational ledger allocated to one independent quadratic degree of freedom. By Lemma 3.19, every active quadratic DOF receives exactly this amount l.

- **Square amplitude on S^1 (1 DOF):** $\beta^2 = 1 \cdot l$
- **Square amplitude on S^2 (2 DOF):** $\kappa^2 = 2 \cdot l$
- **Substituting l immediately yields closure:** $\kappa^2 = 2\beta^2$

Any non-quadratic exchange law, same as any exchange rate other than 2 would require introduction of unjustified ontology either unequal weighting of degrees of freedom violating DOF-Indifference Lemma (3.19) or an additional free parameters violating all four principles (2.2, 2.3, 2.4, 2.1). Therefore:

$$\underbrace{1\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\text{Amplitude}^2}_{\text{on } S^1}$$

□

Remark 3.21 (Carrier capacities fix the rate). *The exchange rate* 2 *exactly mirrors the ratio of the carriers' maximum absolute capacities. The one-DOF carrier* S^1 *saturates at* $\beta^2_{\max} = 1$ *(light,* $\beta_Y = 0$*). The two-DOF carrier* S^2 *saturates at* $\kappa^2_{\max} = 2$ *(the combined extremal-Kerr state* $r = \frac{1}{2}R_s$, $\beta = 1$ *(R.O.M. sec:Kerr)). Equal quadratic weight per DOF (Lemma 3.19) guarantees that the active ledger ratio (*$\kappa^2/\beta^2 = 2$*) is identical to the absolute capacity ratio (*$\kappa^2_{\max}/\beta^2_{\max} = 2$*).*

Definition 3.22 (Closure Factor).

$$\delta \equiv \frac{\kappa_o^2}{2\beta_o^2}$$

Here and throughout all RG papers κ, β *are global and* κ_o, β_o *are orbital phase dependant local amplitudes:* $\kappa^2 = \langle \kappa_o^2 \rangle$ *and* $\beta^2 = \langle \beta_o^2 \rangle$ *For circular orbits,* $\delta \equiv 1$*. For elliptic orbits locally,* $\delta_{cycle} \neq 1$*. For bound elliptical orbits,* $\langle \delta \rangle_{cycle} = 1$ *to remain* relationally closed.

Corollary 3.23 (Relational Closure Condition). *Closed systems (momentary or periodic) satisfy* $\kappa^2 = 2\beta^2$*. Open systems display* $\langle \delta \rangle_{cycle} \neq 1$*, the magnitude of which quantifies the energy flow through unaccounted channels. When all channels are included, closure is restored.*

Illustrative Examples.

- **Circular Orbit (Closed).** A body at any orbital phase satisfies $\kappa_o^2 = 2\beta_o^2$. The entire active ledger is partitioned identically between kinetic and gravitational projections; no internal geometric oscillation and no external channel exists.
- **Elliptical Orbit (Closed).** A body satisfies $\langle \kappa_o^2 \rangle = 2\langle \beta_o^2 \rangle$ as an average per orbital cycle due to the internal geometric oscillation ("breathing") of eccentric systems. This internal oscillation is restricted by the Energy-Symmetry Law (3.11.3) so that the difference $\beta^2 = \kappa_o^2 - \beta_o^2 =$ constant at any orbital phase (R.O.M. sec:vis-viva). No external channel exists.
- **Radiating Binary (Open).** An elliptical compact binary violates $\langle \kappa_o^2 \rangle = 2\langle \beta_o^2 \rangle$ when only orbital degrees of freedom are counted. The closure defect δ quantifies the relational ledger lost to gravitational radiation. Including all channels restores closure.

Remark 3.24 (Geometric Origin of Physical Law). *The relation between kinetic and potential energy projections is a geometric necessity of closure. Aspects of WILL maintain coherence through the invariant DOF ratio. This is the relational origin of the* ***virial theorem****:*

$$\boxed{\textit{Geometric Distribution } (\kappa^2) \;\equiv\; 2 \times \textit{Kinetic Distribution } (\beta^2).}$$

Structural Independence of the Force Law and Virial Coefficient

In classical mechanics, the virial coefficient is dynamically coupled to the spatial force law. For a power-law potential $V \propto r^n$, the general virial theorem dictates $2\langle T \rangle = (n+1)\langle V \rangle$. For Newtonian gravity ($n = -1$), this yields the coefficient 2. In this legacy view, the force law dictates the virial coefficient; they are two sides of the same dynamical coin.
WILL Relational Geometry dismantles this coupling. The $1/r$ proportionality and the virial coefficient 2 are proven to emerge from **entirely independent geometric channels**:

- **Channel 1 (Potential–Distance):** The $\kappa^2 \propto 1/r$ relation is derived strictly from the operational definition of spatial distance and the S^2 carrier's saturation geometry. The S^1 carrier and DOF ratios are entirely absent from this derivation.
- **Channel 2 (Potential–Kinetic Exchange):** The $\kappa^2 = 2\beta^2$ relation is derived strictly from the topological DOF ratio (1 for S^1, 2 for S^2) enforced by DOF-Indifference. Spatial distance r is entirely absent from this derivation.

Because both channels derive from the same underlying carrier topology, their mutual consistency is guaranteed *geometrically*, not dynamically. WILL precludes the possibility of a $1/r$ gravitational potential exhibiting a virial coefficient other than 2. This structural identity replaces the need for postulated equations of motion, Lagrangians, or action principles as foundational primitives.

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3.11 Relational Shift and Energy Symmetry

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Q as Total Relational Shift

3.11.1 Total Relational Shift Q and Reciprocity

When an observer targets another system, the assigned Total Relational Shift norm Q is defined by the sum of squared amplitudes:

$$\boxed{Q^2 = \beta^2 + \kappa^2} \tag{17}$$

Remark 3.25 (Closure-specific simplification)**.** *Under closure $\kappa^2 = 2\beta^2$ (circular/periodic systems), the norm reduces to $Q^2 = 3\beta^2$. In general (open or local elliptic) configurations, the full definition with orbital phase "O" dependence $Q_o(O)^2 = \beta_o(O)^2 + \kappa_o(O)^2$ must be used. See R.O.M. addition for details.*

Theorem 3.26 (Relational Reciprocity)**.** *The operation of self-centering $(\beta, \kappa) = (0, 0)$ is a strict geometric necessity of relational origin principle (2.2), causal balance, and the Total Relational Shift Q between two systems is invariant regardless of the observing frame:*

$$Q^2_{A\to B} = Q^2_{B\to A}.$$

Proof. Consider two interacting systems, A and B. By relational origin principle (2.2) for observer A to measure the relational shift of B, system A must serve as the zero-state reference of its own local frame . This geometrically forces the self-centering operation:

$$(\beta_A, \kappa_A) = (0, 0).$$

From this relational origin, A measures the projections of B as (β_B, κ_B), yielding the norm:

$$Q^2_{A\to B} = \beta_B^2 + \kappa_B^2.$$

Conversely, when B observes A, system B constitutes the local zero-state $(\beta_B, \kappa_B) = (0, 0)$. This forces B to measure the exact same total shift norm for A:

$$Q^2_{B\to A} = \beta_A^2 + \kappa_A^2 = Q^2_{A\to B}.$$

Therefore, reciprocity is not a vector symmetry mapped onto a shared background space. It is the strict invariance of the relational norm Q under the self-centering operation required by relational origin principle (2.2). □

3.11.2 Relational Reciprocity and Self-Centering

Summary

$$\boxed{\text{Relational reciprocity} \;=\; \text{invariance of the norm } Q \text{ under self–centering.}}$$

There is no common background arena. There are only mutual total shift magnitudes Q computed from each observer's own relational origin.

3.11.3 Causal Continuity and Energy Symmetry

Theorem 3.27 (Energy Symmetry)**.** *Any closed system conserves its relational energy ledger with an invariant zero sum of total transformations. Transitions between states of observers (per unit of rest energy) must always sum to zero. Nothing is created, nothing is destroyed:*

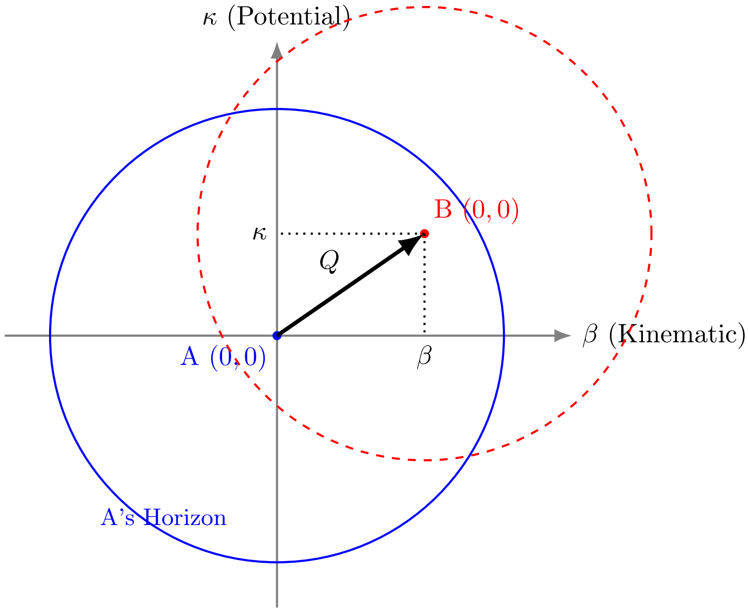
$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0. \tag{18}$$

Proof. By Theorems **Relational Closure (2.5) and Relational Invariance** (2.6), the total local energy of a system is determined by the ratio of the potential phase to the kinematic phase. The total local energy is:

$$E_{\text{loc}} = E_0 \frac{\kappa_X}{\beta_Y} \tag{19}$$

Consider two observers, A and B. The specific energy difference perceived for a transition between their states is determined by the difference in their total phase scaling:

$$\Delta E_{A\to B} = E_0 \left(\frac{\kappa_{X,B}}{\beta_{Y,B}} - \frac{\kappa_{X,A}}{\beta_{Y,A}}\right) \tag{20}$$



Interaction Condition:
$Q < 1$ (Centers are mutually enclosed)

Figure 1: **Relational Self-Centering.** The total shift Q is defined by the amplitude projections β and κ. Interaction is causal only when the relational origin (0, 0) of B lies within A's horizon ($Q < 1$), ensuring mutual coverage.

Conversely, the specific energy difference perceived for the reverse transition is:

$$\Delta E_{B\to A} = E_0\left(\frac{\kappa_{X,A}}{\beta_{Y,A}} - \frac{\kappa_{X,B}}{\beta_{Y,B}}\right) \tag{21}$$

Summing these transfers gives:

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0 \tag{22}$$

Thus, the Energy-Symmetry Law emerges directly from the phase ratio difference, ensuring causal continuity and unbroken bookkeeping across all transitions. The historical separation of kinetic and potential energy appears as a first-order Taylor approximation of this unified phase ratio. □

Theorem 3.28 (Universal Rate of Change and the Horizon of Causality)**.** *The unit radius of kinetic and potential carriers represent the Constant Universal Rate of Change, the Global Tempo of Transformations. The angles θ_1 and θ_2 control the distribution of this constant rate between the phase and amplitude components. What we perceive as the speed of light or event horizon is the exhaustion of the phase, as the constant rate becomes fully distributed onto the amplitude axis. The universal speed limit ($v \leq c$) and the Schwarzschild limit ($r \geq R_s$) emerge from the requirement of energetic symmetry as natural boundaries of causal connection.*

Proof. The unit radius of the kinetic carrier S^1 and the potential carrier S^2 represents the Constant Universal Rate of Change—the Global Tempo of Transformations. The closure identities $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$ dictate the distribution of this constant rate between the amplitude (external interaction) and the phase (internal existence).

The total local energy ledger is scaled by the ratio of the potential phase to the kinematic phase: $E_{\text{loc}} = E_0 \frac{\kappa_X}{\beta_Y}$. The Energy-Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$ requires that any transition between states possesses a finite, symmetric counter-transition to close the ledger.

$$\lim_{\beta_Y \to 0} E_{\text{loc}} \to \infty \quad \text{and} \quad \lim_{\kappa_X \to 0} E_{\text{loc}} \to 0 \tag{23}$$

Kinematic Phase Exhaustion (S^1): As a system distributes its Global Transformation Tempo increasingly into external kinematic interaction, the amplitude β approaches 1. To maintain the carrier's closure, the internal kinematic phase approaches exhaustion: $\beta_Y \to 0$. The local energy scaling diverges: $E_{\text{loc}} = E_0 \frac{\kappa_X}{0} \to \infty$. To preserve the Energy-Symmetry Law, the balancing reverse transition must supply an infinite relational counter-transfer. Because finite physical systems possess finite relational capacity, an infinite counterpart is structurally unattainable. The ledger cannot be closed. Therefore, $\beta_Y = 0$ constitutes a strict causal horizon—the Universal Rate of Change limit ($v = c$)—where energetic symmetry prohibits further amplitude growth.

Potential Phase Exhaustion (S^2): As a system distributes its Global Tempo increasingly into external potential depth, the amplitude κ approaches 1. The internal potential phase approaches exhaustion: $\kappa_X \to 0$. The local energy scaling collapses: $E_{\text{loc}} = E_0 \frac{0}{\beta_Y} \to 0$. To preserve the Energy-Symmetry Law, any transition across this boundary requires a counterpart with zero energy capacity to balance the ledger. Consequently, no finite, non-zero transformation can symmetrically cross this threshold. The point $\kappa_X = 0$ constitutes the relational causal horizon (the Schwarzschild radius), where the internal phase ledger is fully depleted into external potential amplitude, isolating the system from finite causal interactions.

Thus, the universal speed limit and the Schwarzschild radius emerge identically as the necessary causal horizons of Relational Geometry, enforcing the invariant symmetry of the energy ledger against phase exhaustion. ☐

Summary

Energy Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$ is the universal relational energy bookkeeping. **The Speed of Light and the Schwarzschild radius** are the boundaries beyond which the energy symmetry law breaks down. **Causality** is an **inevitable** feature of **Relational Geometry**.

4 Stage III: Legacy Translation & Empirical Alignment

Logical Chain Summary

↓*Methodological Constraints*↓
Relational: Closure + Conservation + Isotropy Theorems.
↓ *(apply to existing physics)*↓
Two Primitives: $\underbrace{\text{fixed manifold + metric}}_{\text{structure}} + \underbrace{\text{fields + constants}}_{\text{dynamics}}$
↓ *(Ontological reduction by Relational Origin Principle no background allowed)*↓
One Primitive: *WILL* ≡ *SPACE-TIME-ENERGY* ⟹ SPACETIME ≡ ENERGY
↓ *(by Closure + Conservation + Isotropy Theorems derive primal relational carriers)*↓
S^1 **(1-DOF kinematic)** + S^2 **(2-DOF potential)**
↓ *(By Duality of Relation Lemma)*↓
Relational Coservation: ↓ $\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1$ ↓
$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic}}$ on S^1 and $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential}}$ on S^2
↓ *(By Conservation Theorem)*↓
Minkowski interval: $\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} \implies \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$
Schwarzschild interval: $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} \implies \underbrace{c^2 d\tau^2 = c^2 \left(1 - \frac{R_s}{r}\right) dt^2}_{\text{mathematical scaffolding}}$
↓ *(By DOF-Indifference Lemma: equal quadratic weight to each independent DOF)*↓
Closure Theorem: $\underbrace{1\,\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\,\text{Amplitude}^2}_{\text{on } S^1} \equiv \kappa^2 = 2\beta^2$
↓ *(By Relational Closure and Invariance theorems)*↓
Energy Symmetry Law: $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$
$\Delta E_{A\to B} = E_0 \left(\frac{\kappa_{X,B}}{\beta_{Y,B}} - \frac{\kappa_{X,A}}{\beta_{Y,A}}\right)$; $\Delta E_{B\to A} = E_0 \left(\frac{\kappa_{X,A}}{\beta_{Y,A}} - \frac{\kappa_{X,B}}{\beta_{Y,B}}\right)$
↓ *(check the requaered energy to close the leger at boundaries)*↓
Speed of light: $\beta \to 1 \to \beta_Y \to 0 \to E_{\text{loc}} = E_0 \frac{\kappa_X}{0} \to \infty \to$ **requaers** ∞ **energy;**
Event Horizon: $\kappa \to 1 \to \kappa_X \to 0 \to E_{\text{loc}} = E_0 \frac{0}{\beta_Y} \to 0 \to$ **requaers** 0 **energy.**

$\theta_1 = \arccos(\beta)$, $\theta_2 = \arcsin(\kappa)$, $\kappa^2 = 2\beta^2$, $a =$ semi-major axis	
Algebraic Form	**Trigonometric Form**
$\beta = v/c$	$\beta = \cos(\theta_1)$
$\kappa = \sqrt{R_s/a}$	$\kappa = \sin(\theta_2)$
$\beta_Y = \sqrt{1-\beta^2}$	$\beta_Y = \sin(\theta_1)$
$\kappa_X = \sqrt{1-\kappa^2}$	$\kappa_X = \cos(\theta_2)$
$p = E_0/c \cdot \beta/\beta_Y$	$p = E_0/c \cdot \cot(\theta_1)$
$p_\kappa = E_0/c \cdot \kappa/\kappa_X$	$p_\kappa = E_0/c \cdot \tan(\theta_2)$
$\tau = \beta_Y \kappa_X$	$\tau = \sin(\theta_1)\cos(\theta_2)$
$\tau_Y = \sqrt{1-\tau^2}$	$\tau_Y = \sqrt{1-\tau^2}$
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{3}\beta$	$Q = \sqrt{3}\cos(\theta_1)$
$Q_Y = \sqrt{1-Q^2} = \sqrt{1-\kappa^2-\beta^2} = \sqrt{1-3\beta^2}$	$Q_Y = \sqrt{1-3\cos^2(\theta_1)}$

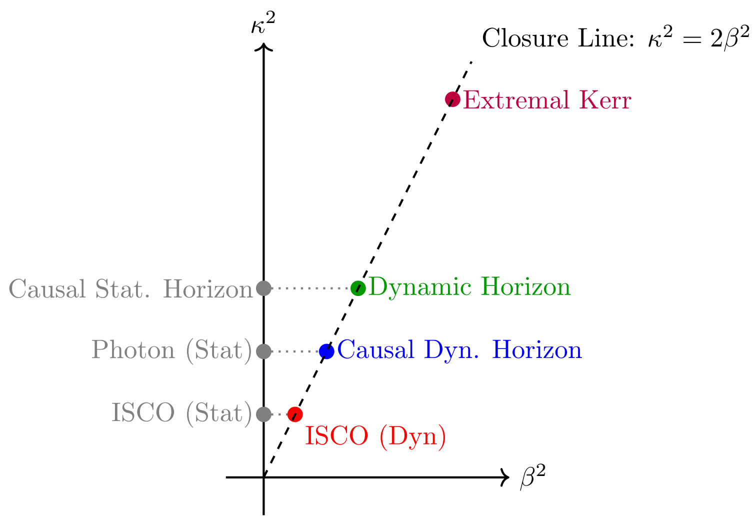
Table 2: Unified representation of relativistic and gravitational effects for closed systems.

Summary

The familiar SR and GR factors emerge here as projections of the same conserved resource. Relativistic (β) and gravitational (κ) modes are not separate "effects" but dual aspects of one energy-transformation constraint revealing their unified origin.

Phenomenon	**Radius r**	β^2	κ^2	Q^2	**Comment**
ISCO (Static location)	$3R_s$	0	$\frac{1}{3}$	$\frac{1}{3}$	Pure coordinate location, $r = R_s/\kappa^2$
ISCO (Dynamic state)	$3R_s$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	Marginal orbital stability, $Q^2 = Q_Y^2 = 1/2$
Photon sphere (Static)	$\frac{3}{2}R_s$	0	$\frac{2}{3}$	$\frac{2}{3}$	Pure coordinate location
Causal Dynamic horizon	$\frac{3}{2}R_s$	$\frac{1}{3}$	$\frac{2}{3}$	1	Relational horizon under closure $\kappa^2 = 2\beta^2$, $Q = 1$
Causal Static horizon	R_s	0	1	1	Purely gravitational closure limit, $\kappa^2 = 1$
Dynamic horizon	R_s	$\frac{1}{2}$	1	$\frac{3}{2}$	Orbital closure at Schwarzschild radius
Extremal Kerr horizon	$\frac{1}{2}R_s$	1	2	3	Maximal rotation, $\beta = 1$, merged horizons

Table 3: Critical radii and their projectional parameters. Static coordinate locations ($\beta = 0$) are strictly distinguished from their corresponding dynamic orbital states that emerge from the closure law $\kappa^2 = 2\beta^2$.



4.1 Equivalence Principle as Derived Identity

Lemma 4.1 (Unified Relational Scaling)**.** *Within the relational framework of WILL, both kinematic (S^1) and gravitational (S^2) transformations act as projections of the same invariant energy E_0. The total local energy scale is determined by the ratio of the potential phase to the kinematic phase:*

$$E_{\text{loc}} = E_0 \frac{\kappa_X}{\beta_Y} = E_0 \sqrt{\frac{1-\kappa^2}{1-\beta^2}}.$$

Proof. On the kinematic circle S^1, the internal phase corresponds to $\beta_Y = \sin\theta_1$. On the gravitational sphere S^2, the internal phase is $\kappa_X = \cos\theta_2$. The total local energy is defined by the ratio of these internal phases, scaling the invariant rest energy E_0. This scaling preserves the closure identity of the respective carriers and governs all external manifestations of the system's energy. □

Theorem 4.2 (Equivalence of Inertial and Gravitational Response)**.** *The total local energy scale is given by the phase ratio:*

$$E_{\text{loc}} = E_0 \frac{\kappa_X}{\beta_Y} = E_0 \sqrt{\frac{1-\kappa^2}{1-\beta^2}}.$$

The corresponding inertial and gravitational projections share a single operational factor,

$$\tilde{p} = \frac{E_{\text{loc}}}{c}\beta, \qquad \tilde{p}_\kappa = \frac{E_{\text{loc}}}{c}\kappa,$$

both governed by the same effective mass

$$m_{\text{eff}} = \frac{E_0 \kappa_X}{\beta_Y c^2} = \frac{E_{\text{loc}}}{c^2}.$$

Therefore,

$$\boxed{m_g \equiv m_i \equiv m_{\text{eff}}},$$

and the Einstein equivalence principle follows as a necessary structural identity of WILL.

Corollary 4.3 (Mass Invariance under Relational Scaling)**.** *The invariant core E_0 denotes the complete internal equilibrium state ($\beta_Y = \kappa_X = 1$). Relational factors β_Y and κ_X rescale only external manifestations (energy, momentum, and rates), while E_0 remains unchanged. Hence,*

$$\boxed{m_g \equiv m_i \equiv m = E_0/c^2},$$

is the definition of rest invariance itself.

Remark 4.4 (Composition-Independence)**.** *Decomposing the invariant rest energy into internal channels,*

$$E_0 = \sum_a E_0^{(a)},$$

each term couples identically through the same phase ratio scaling:

$$E_{\text{loc}} = \sum_a \left(E_0^{(a)} \frac{\kappa_X}{\beta_Y} \right).$$

Since all channels scale by the same phase ratio $\frac{\kappa_X}{\beta_Y}$, ratios between channels cancel in all observables. Therefore, composition-independence of motion (Eotvos universality) follows identically, without requiring a postulate $m_g = m_i$.

Remark 4.5 (Quantum Interface)**.** *The relational phase increment inherits the same scaling:*

$$\Delta\phi \propto E_{\text{loc}}\, \Delta\lambda,$$

where $\Delta\lambda$ is the internal ordering parameter. Thus both kinematic and gravitational phase shifts share the same phase ratio scaling $\frac{\kappa_X}{\beta_Y}$, yielding composition-independent matter-wave interference patterns.

Summary:

In WILL, the equivalence of inertial and gravitational mass follows necessarily from the phase ratio scaling of the relational geometry. What General Relativity posits as a postulate, WILL reveals as a corollary.

4.2 Classical Mechanics

4.2.1 The Phase Scaling and the Classical $\frac{1}{2}$

The total energy in RG is given by the phase ratio:

$$\frac{E_{\text{tot}}}{E_0} = \frac{\kappa_X}{\beta_Y} = \frac{\sqrt{1-\kappa^2}}{\sqrt{1-\beta^2}} = (1-\kappa^2)^{1/2}(1-\beta^2)^{-1/2} \tag{24}$$

In the weak-field, low-velocity regime ($\beta \ll 1$, $\kappa \ll 1$), the Taylor expansion of the phase scaling yields:

$$\frac{E_{\text{tot}}}{E_0} \approx \left(1 - \frac{1}{2}\kappa^2\right)\left(1 + \frac{1}{2}\beta^2\right) \approx 1 - \frac{1}{2}\kappa^2 + \frac{1}{2}\beta^2 \tag{25}$$

The excess energy (the kinetic and potential ledgers) per unit rest energy is therefore:

$$\frac{E_{\text{tot}} - E_0}{E_0} \approx \frac{1}{2}\beta^2 - \frac{1}{2}\kappa^2 \tag{26}$$

Relational Origin of the Classical $\frac{1}{2}$

The historical factor $\frac{1}{2}$ in classical kinetic and potential energy formulas is the first-order Taylor coefficient of the phase ratio κ_X/β_Y. Newtonian physics emerges as the linear approximation of the relational phase ratio. Classical mechanics captures the first-order, a testament to its empirical adequacy in the weak-field regime.

4.2.2 Physical Interpretation

This demonstrates that the Keplerian total energy is a direct linear projection of a deeper relational structure. The difference of squared amplitudes ($\beta^2 - \kappa^2$) emerges invariantly from the closed geometry of the carriers. The structure of the energy relation is a direct consequence of the relational closure of S^1 and S^2.

4.3 Two-Point Origin of the Lagrangian and Hamiltonian

In the standard formulation of mechanics, the Lagrangian $L = T - V$ and the Hamiltonian $H = T + V$ are treated as fundamental functions of a single configuration point $\mathbf{r}$. Relational Geometry reveals that both are linearized limits of a deeper two-point energy balance: the **Energy-Symmetry Law**

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0,$$

where the specific energy (per unit rest energy) perceived by an observer at state A for a transition to state B is determined by the difference in their total phase scaling:

$$\boxed{\Delta E_{A\to B} = \left(\frac{\kappa_X}{\beta_Y}\right)_B - \left(\frac{\kappa_X}{\beta_Y}\right)_A}. \tag{27}$$

Here $\beta = v/c$ is the kinematic projection on the carrier S^1 and $\kappa^2 = R_s/r$ is the potential projection on the carrier S^2.

4.3.1 The linearized relational limit

To connect with legacy mechanics, we examine the weak-field, low-velocity regime ($\beta \ll 1, \kappa \ll 1$). The Taylor expansion of the phase ratio yields $\frac{\kappa_X}{\beta_Y} \approx 1 - \frac{1}{2}\kappa^2 + \frac{1}{2}\beta^2$. Substituting this into the two-point law (Eq. 27) gives:

$$\Delta E_{A\to B} \approx \left(1 - \tfrac{1}{2}\kappa_B^2 + \tfrac{1}{2}\beta_B^2\right) - \left(1 - \tfrac{1}{2}\kappa_A^2 + \tfrac{1}{2}\beta_A^2\right).$$

The constant 1 cancels out, leaving the linearized specific energy transfer:

$$\Delta E_{A\to B} \approx \tfrac{1}{2}\left(\kappa_A^2 - \kappa_B^2\right) + \tfrac{1}{2}\left(\beta_B^2 - \beta_A^2\right). \tag{28}$$

The factor $\frac{1}{2}$ appears here as the first-order Taylor coefficient of the inverse-phase scaling.

4.3.2 First ontological collapse: the Newtonian Hamiltonian

The standard Newtonian Hamiltonian is obtained by the first ontological violation. By the Principle of Relational Reciprocity (3.26), an observer self-centers at their own relational origin, meaning observer A measures their own state as $\kappa_A = 0$ and $\beta_A = 0$. (Legacy physics interprets this self-centered zero-state as placing the reference frame at "infinity").

Applying this self-centered zero-state to the linearized two-point law (Eq. 28), the transition collapses into a property of a single state B:

$$E \approx \tfrac{1}{2}\beta_B^2 - \tfrac{1}{2}\kappa_B^2.$$

Multiplying by mc^2 and substituting the legacy definitions of β and κ gives:

$$H \;=\; \tfrac{1}{2}mv^2 \;-\; \frac{GMm}{r},$$

which is the Newtonian Hamiltonian. The potential information, which belongs to the two-point difference, is artificially reassigned to a single point via an externally postulated potential function $V(r) = -GMm/r$.

4.3.3 Second ontological collapse: the Newtonian Lagrangian

If one starts from the same linearized two-point law but enforces the second ontological violation — forcing the two distinct points to occupy the same spatial coordinate, $r_A = r_B = r$ — the relational structure collapses further. Setting $\kappa_A^2 = \kappa^2(r)$ and $\beta_A = 0$ while keeping $\beta_B = \beta$, the linearized equation becomes:

$$\Delta E_{A\to B} \approx \tfrac{1}{2}\big(\kappa^2(r) - \kappa^2(r)\big) + \tfrac{1}{2}\beta^2 = \tfrac{1}{2}\beta^2.$$

The potential information completely vanishes from the relational ledger. To recover a dynamical equation, legacy mechanics must separately postulate an external potential energy function $V(r)$ and add it ad-hoc to the kinetic term. Assigning the opposite sign convention to this ad-hoc potential yields the Newtonian Lagrangian:

$$L \;=\; \tfrac{1}{2}mv^2 \;+\; \frac{GMm}{r}.$$

Both the Hamiltonian and the Lagrangian are revealed as single-point artifacts of collapsing the true two-point phase ratio law.

4.3.4 Summary

True Two-point relational form	Collapsed single-point form
$\Delta E_{A\to B} = \left(\dfrac{\kappa_X}{\beta_Y}\right)_B - \left(\dfrac{\kappa_X}{\beta_Y}\right)_A$	$L = \tfrac{1}{2}mv^2 + \dfrac{GMm}{r}$ $H = \tfrac{1}{2}mv^2 - \dfrac{GMm}{r}$
The collapse is performed by Taylor linearizing the phase ratio (introducing the $\frac{1}{2}$), applying the self-centered origin (Hamiltonian), and identifying $r_A = r_B$ (Lagrangian). The two-point phase ratio contains the full energy balance without the need for a separate potential.	

Table 4: True two-point relational law versus collapsed single-point forms.

Key insight

The Lagrangian and Hamiltonian are linearized artifacts of the single-point collapse of the two-point Energy-Symmetry Law. In the relational framework, all dynamical information is encoded in the dimensionless phase ratio κ_X/β_Y and its geometric closure. No potential function, no force law, and no action principle are required as foundational primitives.

Remark 4.6 (Mathematical Status: Groupoid vs. Group)**.** *From a category-theoretic perspective, the relational transitions $\Delta E_{A\to B}$ form a* ***Groupoid*** *structure, where operations are defined only between specific connected states. Standard field theories (Hamiltonian/Lagrangian) rely on the collapse of this structure into a* ***Group*** *acting on a global manifold (quotienting the state space). Thus, the relational Energy-Symmetry Law operates at the Groupoid level, prior to the reduction into global field theories.*

Legacy Dictionary (for conventional formalisms).

Within RG, all physical content is expressed purely in terms of relational projections β and κ on S^1 and S^2. For readers accustomed to standard frameworks, the following translation rules apply:

1. *General Relativity (metric form):*

$$\kappa_X \mathrel{\hat{=}} \sqrt{-g_{tt}} \quad \text{(static spacetimes)}, \qquad \beta \mathrel{\hat{=}} \frac{\|u^\mu_{\text{spatial}}\|}{u^t c}.$$

2. *Canonical mechanics (Lagrangian/Hamiltonian):* Quantities such as $p_i = \partial L/\partial \dot{q}^i$ do not belong to the ontology of RG. They are secondary derived quantities arising only after collapsing the two-point relational structure into a local single-point formalism.

Here the symbol $\hat{=}$ denotes a pragmatic dictionary entry for translation into legacy notation, rather than an ontological identity.

4.3.5 Third Ontological Collapse: Newton's Third Law

We now demonstrate that Newton's Third Law, like the Lagrangian and Hamiltonian, emerges as a limiting case of RG. It arises as a necessary mathematical consequence of the same ontological collapse — forcing a two-point relational law into a single-point, instantaneous formalism.

Theorem 4.7 (Newton's Third Law as a Single-Point Limit)**.** *The Energy–Symmetry Law ($\Delta E_{A\to B} + \Delta E_{B\to A} = 0$) mathematically necessitates Newton's Third Law ($\vec{F}_{AB} = -\vec{F}_{BA}$) in the classical, non-relativistic limit where the two-point relational energy budget is collapsed into a single-point potential function $U(\vec{r})$.*

Proof. We begin with the Energy–Symmetry Law (Section 3.11.3), the principle of causal balance for state transitions:

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0.$$

In the classical limit, this two-point relational law is ontologically collapsed into a single-point potential energy function, U. This function is assumed to depend only on the relative positions of the two interacting entities, A and B:

$$U = U(\vec{r}) \quad \text{where} \quad \vec{r} = \vec{r}_B - \vec{r}_A.$$

This $U(\vec{r})$ is the classical approximation of the system's relational energy ledger. In this collapsed formalism, the force $\vec{F}$ is defined as the negative gradient of this potential.

(1) Force on B by A ($\vec{F}_{AB}$): This force is found by taking the gradient with respect to B's coordinates:

$$\begin{aligned}
\vec{F}_{AB} &= -\nabla_B U(\vec{r}_B - \vec{r}_A) && (29)\\
&= -\frac{dU}{d\vec{r}} \cdot \frac{\partial \vec{r}}{\partial \vec{r}_B} && (30)\\
&= -\nabla U(\vec{r}) \cdot \mathbf{I} && (31)\\
&= -\nabla U(\vec{r}) && (32)
\end{aligned}$$

(2) Force on A by B ($\vec{F}_{BA}$): This force is found by taking the gradient with respect to A's coordinates:

$$\begin{aligned}
\vec{F}_{BA} &= -\nabla_A U(\vec{r}_B - \vec{r}_A) && (33)\\
&= -\frac{dU}{d\vec{r}} \cdot \frac{\partial \vec{r}}{\partial \vec{r}_A} && (34)\\
&= -\nabla U(\vec{r}) \cdot (-\mathbf{I}) && (35)\\
&= +\nabla U(\vec{r}) && (36)
\end{aligned}$$

(3) Conclusion: By direct comparison of the results, we find:

$$\vec{F}_{AB} = -\nabla U(\vec{r}) \quad \text{and} \quad \vec{F}_{BA} = +\nabla U(\vec{r}).$$

Therefore, within the single-point collapsed formalism:

$$\boxed{\vec{F}_{AB} = -\vec{F}_{BA}}$$

This completes the proof. Since all physical observations are relational (2.2), embedding the energy budget within an absolute spatial container ($\vec{r}_A, \vec{r}_B$) introduces unobservable parameters, violating Epistemic Hygiene (2.1). To remain empirically grounded, the collapsed potential U must depend exclusively on the measurable physical relation $\vec{r}_B - \vec{r}_A$. Consequently, the law of equal and opposite forces emerges as the mathematical signature of any causally bound, background-independent system $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$. □

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4.3.6 Fourth Ontological Collapse: The Einstein-Hilbert Action (S_{EH})

The Einstein-Hilbert action ($S_{EH} = \frac{1}{16\pi G}\int R\sqrt{-g}\,d^4x$) represents the final reduction of relational connections between states into localized coordinate points. Standard tensor calculus employs the limit $dx \to 0$ to compress relational tension into the local Ricci scalar (R). This operation necessitates the postulation of a smooth background manifold and the application of the Principle of Stationary Action ($\delta S_{EH} = 0$) to maintain global equilibrium across arbitrary coordinates.

4.3.7 Components Mapping

- **Global 4D Volume** ($\int\sqrt{-g}\,d^4x \to 1$): Instead of an integration over a background manifold, the total relational capacity in RG is bound by the unitary closure of the carriers: $\kappa^2 + \kappa_X^2 = 1$ and $\beta^2 + \beta_Y^2 = 1$ (3.6).
- **The Ricci Scalar** ($R \to \kappa^2, \beta^2$): The differential approximation of spacetime curvature is replaced directly by the finite, dimensionless relational projections.
- **The Variational Search** ($\delta S = 0 \to \Delta E_{A\to B} + \Delta E_{B\to A} = 0$): The continuous minimization of the action integral is the continuum shadow of the Energy-Symmetry Law. Requiring $\delta S = 0$ over a manifold is mathematically equivalent to requiring $\Delta E = 0$ for all local transitions, ensuring the global relational ledger balances perfectly.
- **The full mapping is provided in R.O.M. addition.**

Empirical Grounding

The empirical necessity of the action (S_{EH}) manifests in phenomena such as gravitational light deflection, which legacy physics derives via the addition of a curved spatial metric (g_{rr}). RG achieves the identical empirical result without introducing new ontological primitives, deriving it directly from the S^2 carrier geometry: (Gravitational Deflection and Lensing).

4.4 Earth–GPS System: Standard GR as First-Order Approximation of RG

We made lots of derivations and daring claims. Its time to put them to the test. For our first tests we will use the standard Earth - GPS satellite system to challenge theorems and lemmas: (2.5; 2.7; 3.3; 3.6; 3.7; 3.10; ??; 3.12; 3.20; 3.11.3; 5.2; 4.3; 4.3.5)

4.4.1 Statement of the Verification Problem

The standard post-Newtonian expression for the secular time shift between a clock on the Earth's surface and a clock on a circular GPS orbit, accumulated over one solar day, is

$$\Delta t_{\mathrm{GR}} = \left(\frac{\Phi_{\mathrm{GPS}} - \Phi_E}{c^2} - \frac{v_{\mathrm{GPS}}^2 - v_E^2}{2\,c^2}\right) D\,M, \tag{37}$$

with $D = 86{,}400\,\mathrm{s}$, $M = 10^6$, $\Phi(r) = -GM_\oplus/r$, $v_E = 2\pi R_E/D$, and $v_{\mathrm{GPS}} = \sqrt{GM_\oplus/r_{\mathrm{GPS}}}$.

WILL Relational Geometry (WILL RG) expresses the same time shift as an exact algebraic ratio of two-carrier projection invariants $\tau \equiv \sqrt{1-\kappa^2}\,\sqrt{1-\beta^2}$:

$$\Delta t_{\mathrm{RG}} = \left(1 - \frac{\sqrt{1-\kappa_E^2}\,\sqrt{1-\beta_E^2}}{\sqrt{1-\kappa_{\mathrm{GPS}}^2}\,\sqrt{1-\beta_{\mathrm{GPS}}^2}}\right) D\,M, \tag{38}$$

where $\kappa^2(r) = 2GM_\oplus/(rc^2)$, $\beta^2 = v^2/c^2$, and the orbital closure condition $\beta_{\mathrm{GPS}}^2 = \kappa_{\mathrm{GPS}}^2/2$ holds for circular motion.

The question addressed in this section is whether (37) is the leading-order Taylor expansion of (38) in the small parameters κ^2 and β^2. If so, the algebraic difference of the two coefficients must vanish identically at first order, and the residual at finite values of the small parameters must reproduce the analytically derived second-order correction.

4.4.2 Methodology

The verification is split into two independent parts, both executed in a single public notebook (Section 4.4.5).

Part I — symbolic identity. Introduce the dimensionless geometric ratio $\rho \equiv R_E/r_{\mathrm{GPS}}$, which is treated as a finite (order-unity) parameter and is *not* expanded. The exact WILL RG shift coefficient is then

$$\delta_{\mathrm{RG}}(\kappa_E^2, \beta_E^2, \rho) \;=\; 1 - \frac{\sqrt{(1-\kappa_E^2)(1-\beta_E^2)}}{\sqrt{(1-\kappa_E^2\rho)(1-\kappa_E^2\rho/2)}}. \tag{39}$$

A single bookkeeping parameter ε is introduced via $\kappa_E^2 \to \varepsilon\,\kappa_E^2$, $\beta_E^2 \to \varepsilon\,\beta_E^2$, and the Taylor series of (39) is computed to second order in ε using `sympy.series`. The first-order coefficient $\delta_{\mathrm{RG}}^{(1)}$ is then subtracted from the dimensionless form of the GR coefficient δ_{GR} obtained from (37) under the substitutions $\Phi/c^2 = -\kappa^2/2$ and $v^2/c^2 = \beta^2$. The result is simplified algebraically.

Part II — numerical comparison. Both (37) and (38) are evaluated with 40-digit precision arithmetic (`mpmath`, `mp.dps = 40`) using a single set of input parameters: $c = 299{,}792{,}458\,\mathrm{m\,s^{-1}}$, $GM_\oplus = 3.986004418\times10^{14}\,\mathrm{m^3\,s^{-2}}$, $R_E = 6{,}378{,}137\,\mathrm{m}$, $r_{\mathrm{GPS}} = 26{,}561{,}750\,\mathrm{m}$, $D = 86{,}400\,\mathrm{s}$. The closure residual $\beta_{\mathrm{GPS}}^2 - \kappa_{\mathrm{GPS}}^2/2$ is monitored as a numerical consistency check. The numerical discrepancy $\Delta t_{\mathrm{RG}} - \Delta t_{\mathrm{GR}}$ is then compared with the analytically predicted second-order correction $\delta_{\mathrm{RG}}^{(2)}\,D\,M$ obtained from Part I.

4.4.3 Results

Symbolic. The first-order Taylor coefficient of the WILL RG shift, expressed in κ_E^2, β_E^2, and the geometric ratio ρ, is

$$\delta_{\mathrm{RG}}^{(1)} \;=\; \frac{\kappa_E^2}{2} + \frac{\beta_E^2}{2} - \frac{3\,\kappa_E^2\,\rho}{4}. \tag{40}$$

The GR coefficient, expressed in the same variables, is

$$\delta_{\mathrm{GR}} \;=\; \frac{\kappa_E^2}{2} + \frac{\beta_E^2}{2} - \frac{3\,\kappa_E^2\,\rho}{4}. \tag{41}$$

The algebraic difference $\delta_{\mathrm{RG}}^{(1)} - \delta_{\mathrm{GR}}$ vanishes identically.

The second-order content of (39), which is absent from the additive GR formula (37), is

$$\delta_{\mathrm{RG}}^{(2)} \;=\; \frac{\kappa_E^4}{8} + \frac{\beta_E^4}{8} - \frac{\beta_E^2\,\kappa_E^2}{4} + \frac{3\,\kappa_E^4\,\rho}{8} - \frac{19\,\kappa_E^4\,\rho^2}{32} + \frac{3\,\beta_E^2\,\kappa_E^2\,\rho}{8}. \tag{42}$$

Three structural features of (42) have no analogue in any rearrangement of the additive GR sum (37): (i) the pure quadratic contributions $\kappa_E^4/8$ and $\beta_E^4/8$ arising from higher-order expansion of $\sqrt{1-x}$; (ii) the cross-term $-\beta_E^2\kappa_E^2/4$, which originates from the multiplicative coupling of the gravitational and kinematic projection factors within $\tau_E/\tau_{\mathrm{GPS}}$; and (iii) the ρ^2 contribution, which encodes nonlinear dependence on the relative depth of the gravitational well.

Numerical. With the parameter set listed above:

quantity	value [μs day^{-1}]
Δt_{RG} (exact)	38.5421472752
Δt_{GR} (first order)	38.5421472451
$\Delta t_{\mathrm{RG}} - \Delta t_{\mathrm{GR}}$	3.017×10^{-8}
$\delta_{\mathrm{RG}}^{(2)}\cdot D\,M$ (predicted)	3.017×10^{-8}
residual	$\sim 10^{-17}$

The numerical discrepancy $\Delta t_{\mathrm{RG}} - \Delta t_{\mathrm{GR}}$ matches the analytically derived second-order correction $\delta_{\mathrm{RG}}^{(2)}\,D\,M$ to within the limit set by the 40-digit floating-point arithmetic. The closure residual $\beta_{\mathrm{GPS}}^2 - \kappa_{\mathrm{GPS}}^2/2$ vanishes to $\sim 10^{-50}$. The relative magnitude of the term omitted by the GR additive formula is $\sim 7.8\times10^{-10}$ of the total shift.

4.4.4 Interpretation

Equations (40)–(41) establish that the additive post-Newtonian formula (37) coincides exactly with the linear Taylor coefficient of the WILL RG exact algebraic ratio (38). This is an algebraic identity in $(\kappa_E^2, \beta_E^2, \rho)$, not a numerical near-match: the additive sum-of-corrections structure of (37) is recovered as the unique leading-order content of the multiplicative ratio $\tau_E/\tau_{\mathrm{GPS}}$.

For the Earth–GPS configuration, the second-order term (42) contributes at the level of $\sim 3\times10^{-8}\,\mu\mathrm{s\,day^{-1}}$, i.e. $\sim 8\times10^{-10}$ of the total accumulated shift. This is below the present precision of operational GPS clock comparisons and does not, by itself, constitute a distinguishing observation between the two formulations in this regime. The structural difference between an additive and a multiplicative composition of projection factors — in particular the presence of the $\beta^2\kappa^2$ cross-term in (42) — becomes quantitatively relevant only when κ^2 or β^2 approach order unity.

The scope of the present statement is restricted to the relation between the WILL RG closed-form expression (38) and the *additive post-Newtonian* form of the GR time-dilation formula (37). A comparison against fully non-linear geodesic integration in the exact Schwarzschild geometry combined with special-relativistic kinematics for the rotating Earth surface is a separate computation and is not addressed in this section.

4.4.5 Reproducibility

The complete symbolic and numerical verification reported in this section is contained in a single public Google Colab notebook, comprising the two scripts used to produce (40), (42), and the numerical table above:

Desmos project

Earth-GPS time, energy symmetry and WILL invariant test

Colab Notebook

Earth-GPS 1st order RG=GR.ipynb

The notebook has no external dependencies beyond `sympy` and `mpmath` and is self-contained. All input parameters are listed explicitly in the numerical script; any change in r_{GPS}, R_E, or $GM_\oplus$ rescales both Δt_{RG} and Δt_{GR} consistently and does not affect the symbolic identity $\delta_{\mathrm{RG}}^{(1)} \equiv \delta_{\mathrm{GR}}$.

4.5 Relational Orbital Mechanics (R.O.M.)

Relational Orbital Mechanics (R.O.M.)

Natural evolution of this section within WILL RG Open Research lead it to branch in to its own but not separate R.O.M. paper. It still remain structurally and logically a crucial part of this document that caries most of empirical applications and computational tests.

This closed algebraic system recovers the leading contributions to perihelion precession, gravitational and transverse Doppler shifts, light deflection and frame-dragging through elementary closed-form expressions.

High-precision validation shows that all 66 multi-form identities are internally consistent to 50 significant digits and that the leading terms match the corresponding post-Newtonian expansions of general relativity. Higher-order differences remain below the sensitivity of current observations, including those of the S2 star.

Complete phase-resolved orbits and system scales are recovered directly from spectroscopic and chronometric observables, with G and M entering only as unit-conversion factors. Key features of Kerr geometry follow algebraically from the same closure relation. Gravitational waves and the radiative two-body problem remain to be formulated within the relational framework.

Within R.O.M. we reveal relational origin of:

- Spectroscopic Shifts
- Kepler's Third Law
- Eccentricity
- Vis-Viva
- Classical Acceleration
- Angular Momentum Conservation
- Time-Density Invariant
- Orbital Precession as Phase Accumulation
- Dynamic Event Horizon
- Gravitational Deflection and Lensing
- Schwarzschild metric

- Kerr metric
- Algebraic Determination of Absolute System Scale

We tested this closed algebraical system on:

- GPS-Earth time dilation
- Mercury's Precession and Jupiter's orbit without mass or G
- Earth–Sun L1 Point Distance Derivation
- S2 Star (Sgr A*) 24 Years Data Alignment Comparison: RG vs GR
- Lense-Thirring: LAGEOS-1 Satellite (Earth orbit)

Relational Orbital Mechanics as algebraically closed system of equations developed in this addition demonstrates that the entire phenomenology of motion – emerges as an inevitable algebraic consequence of the Foundational Methodological Principles. All WILL RG results including the complete success of R.O.M. remains an unbroken chain of derivations from first principles under extreme methodological constraints with zero flexibility.

Colab Notebook R.O.M. Full Tests

ROM_FULL_TEST.ipynb - In this notebook we preform 3 wide tests of Relational Orbital Mechanics:

- Algebraically test the closed system (internal consistency of every multi-form identity). All 66 expressions (50-digit, random c,G,M,a,e,O).
- R.O.M. vs GR: We compute the exact Schwarzschild geodesic results by high-precision numerical integration (no approximation), then extract their Taylor coefficients in the SAME small parameter and compare term-by-term to R.O.M..
- ROM reconstructs a, R_s, r_p, r_a and the whole phase-resolved orbit, and recovers GM (a length^3/ time^2 combination, no G) and finally M (kg, G used ONLY here as a units conversion factor).

4.6 Density, Mass, and Pressure

4.6.1 Derivation of Density

Translating RG (2D) to Conventional Density (3D). In RG κ^2 is the 2D parameter defined in the relational carrier S^2 . In conventional physics, the source term is volumetric density ρ, a 3D concept defined by the "cultural artifact" (a Newtonian "cannonball" model) of mass-per-volume .

To compare with measurements that are historically expressed in 3D terms, we need to bridge our 2D theory with 3D units that empirical data is using. We must create a "translation interface". We do this by adopting the conventional (Newtonian) definition of density, $\rho \propto M/r^3$, as our "translation target".

From the projective analysis established in the previous sections:

$$\kappa^2 = \frac{R_s}{r},$$

where κ emerges from the energy projection on the area of unit sphere S^2, and $R_s = 2GM/c^2$ links to the mass scale factor $M = E_0/c^2$.

This leads to mass definition:

$$M = \frac{\kappa^2 c^2 r}{2G}$$

To translate this into a volumetric density, we first adopt the conventional 3D (volumetric) proxy, r^3. This is not a postulate of RG, but the first step in applying the legacy (3D) definition of density:

$$\frac{M}{r^3} = \frac{\kappa^2 c^2}{2Gr^2}$$

This expression, however, is incomplete. Our κ^2 "lives" on the 2D surface S^2 (which corresponds to 4π), while the r^3 proxy assumes a 3D volume (which corresponds to $4\pi/3$). To correctly normalise the 2D parameter κ^2 against the 3D volume, we must apply the geometric normalization factor of the S^2 carrier by dividing by the area of the sphere, which is $1/4\pi$ (we explicitly reject the Newtonian volume integration ($4\pi/3$) in favor of the S^2 carrier's geometric surface (4π). This enforces the relational principle that structural capacity scales exclusively via the 2D spherical boundary of interaction, not by filling an assumed 3D volumetric container).

This normalization is the necessary geometric step to interface the 2D relational carrier (S^2) with the 3D legacy definition of density:

$$\rho = \frac{1}{4\pi}(\frac{\kappa^2 c^2}{2Gr^2})$$
$$\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$$

Local Density $\propto$ Relational Potential Projection

Maximal Density. At $\kappa^2 = 1$ (the horizon condition (for non rotating systems), $r = R_s$), this density reaches its natural bound, $\rho_{\max}$:

$$\rho_{\max} = \frac{c^2}{8\pi G r^2}$$

Normalized Relation. Thus, our "translation" reveals an identity: the potential projection κ^2 is the ratio of density to the maximal density:

$$\boxed{\kappa^2 = \frac{\rho}{\rho_{\max}} \quad \rightarrow \quad \kappa^2 \equiv \Omega}$$

Self-Consistency Requirement

The mass scale factor can be expressed in two equivalent ways.

From the geometric definition:

$$M = \frac{\kappa^2 c^2 r}{2G}.$$

From the energy density:

$$M = \alpha\, r^n\, \rho.$$

Substituting $\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$ into $M = \alpha\, r^n\, \rho$ gives

$$M = \frac{\alpha \kappa^2 c^2 r^{n-2}}{8\pi G}.$$

Equating the two forms:

$$\frac{\alpha r^{n-2}}{8\pi} = \frac{r}{2}.$$

For the mass M to remain a constant independent of the measurement scale r, the exponent must be $n = 3$, yielding $\alpha = 4\pi$. Hence,

$$M = 4\pi r^3 \rho,$$

which closes the consistency loop between the geometric and density-based formulations.

Desmos Project

Derivation of Density

4.6.2 Pressure as Surface Curvature Gradient

In the RG framework pressure is not a thermodynamic assumption but the direct consequence of curvature gradients. The radial balance relation gives

$$P(r) \;=\; \frac{c^4}{8\pi G}\,\frac{1}{r}\,\frac{d\kappa^2}{dr}.$$

Using $\kappa^2 = R_s/r$, one finds $d\kappa^2/dr = -\kappa^2/r$, hence

$$P(r) \;=\; -\,\frac{\kappa^2 c^4}{8\pi G r^2}.$$

Since the local energy density is

$$\rho(r) \;=\; \frac{\kappa^2 c^2}{8\pi G r^2},$$

this yields the invariant equation of state

$$\boxed{P(r) = -\,\rho(r)\, c^2}.$$

Interpretation. P is a surface-like negative pressure (isotropic tension), not a bulk volume pressure. It expresses the resistance of energy-geometry itself to changes in projection.

Consistency. If one formally freezes the projection parameter ($d\kappa^2/dr = 0$), then $P = 0$. But in this case the angular curvature terms remain uncompensated, and the field equation is no longer satisfied. Any nontrivial radial dependence of κ inevitably generates the negative tension

$$P = -\rho c^2,$$

which precisely cancels the residual curvature. Thus the negative pressure is not optional but a necessary ingredient for full self-consistency.

Desmos Project

Derivation of Density and Pressure

Maximum pressure. At the geometric bound $\kappa^2 = 1$ (horizon condition), the density saturates at

$$\rho_{\max} = \frac{c^2}{8\pi G r^2},$$

and the corresponding pressure is

$$P_{\max} = -\,\rho_{\max}\,c^2 \;=\; -\,\frac{c^4}{8\pi G r^2}.$$

This negative surface pressure represents the ultimate tension limit of spacetime fabric at a given scale r.

Pressure in WILL is the intrinsic surface tension of energy-geometry, saturating at $P_{\max} = -c^4/(8\pi G r^2)$.

Physical meaning

The negative pressure is not an exotic substance but the geometric tension required to maintain self-consistency when κ^2 varies radially. It's the relational analogue of surface tension in a soap bubble.

4.7 Unified Geometric Field Equation

From the energy-geometry equivalence, the complete description of gravitational phenomena reduces to a single algebraic relation linking the geometric scale to the energy density ratio:

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho_{field}}{\rho_{max}}}$$

This identity defines the **local energy state of the relational geometry itself**. Here $\rho_{\max} = c^2/(8\pi G r^2)$ is the saturation density limit, and ρ_{field} is the effective energy density of the relational curvature.

4.8 Field Equation and Matter Sources

For a static, spherically symmetric configuration containing matter with density $\rho_{\text{matter}}(r)$, the relationship is governed by the differential accumulation of the potential:

$$\boxed{\frac{d}{dr}(r\,\kappa^2) = \frac{8\pi G}{c^2}\,r^2\rho_{matter}(r)} \tag{43}$$

Remark 4.8 (Scope of the reproduced sector)**.** *The differential form above reproduces the static tt-component. This is not the boundary of the framework: through phase parametrization the same projections describe any single relational orbit — bound at any eccentricity $e = 2\beta_p^2/\kappa_p^2 - 1$, and unbound (deflection, lensing)– and rotating systems (R.O.M. Kerr without metric,). GR direct parameters translational mapping. Current boundaries: sec:challenges.*

The Vacuum Solution ($\rho_{matter} = 0$). In the vacuum region outside a central mass, the source density vanishes ($\rho_{matter} = 0$). The field equation implies conservation of the projection budget:

$$\frac{d}{dr}(r\kappa^2) = 0 \quad \Longrightarrow \quad r\kappa^2 = \text{const} = R_s.$$

Thus, we recover the potential law of WILL RG:

$$\boxed{\kappa^2 = \frac{R_s}{r}.}$$

Resolution of Roles

1. The Identity $\kappa^2 = \rho/\rho_{\max}$ describes the state of the *field* geometry.
2. The Equation $(r\kappa^2)' \sim \rho_{\text{matter}}$ describes how *matter* generates geometry.
In vacuum, the generator is zero, but the field persists as the algebraic structure $\kappa^2 = R_s/r$.

4.9 No Singularities, No Hidden Regions

WILL Relational Geometry resolves the singularity problem by naturally constraining the valid range of radial distance. Minimal radial distance is kept at $r_{min} = R_s/2$ by the fundamental invariance of light speed limit. Parameters stay finite, and energy remains bounded. Black holes become the boundary of relational difference between energy states.

Distance is not a foundational primitive but relational tension between energy states (3.8) The amplitude bound $\kappa^2 \leq 1$ (static case) and $\kappa^2 \leq 2$ (extremal rotation) caps the radial distance by closure theorem (3.10):

No Singularities

$$r = R_s/\kappa^2 \geq R_s/2$$

$$\beta^2_{max} = 1 \Longrightarrow \kappa^2_{max} = 2 \Longrightarrow r_{min} = R_s/\kappa^2_{max} \Longrightarrow r_{min} = R_s/2 \tag{44}$$

The point $r = 0$ is absent from the admissible relational range.
The same bound enforces:

$$\rho \leq 2\rho_{\max} \quad \text{and} \quad |P| \leq |2P_{\max}| = c^4/(8\pi G r^2) \tag{45}$$

Another interesting hypothetical explanation of this phenomenon comes from Relational Orbital Mechanics. Dynamic Relational Horizon provides new point of view on the nature of Black Holes where precession of the perihelium becomes equivalent to orbital frequency $\Delta\varphi = 2\pi$. The orbit starts to fold on itself and nullify any orbital motion. Its not a two-surface enclosing a region — so there is no interior to be hidden in the first place. However the concept of dynamic horizon remain purely mathematical and requires more research.

5 Stage IV Consequences and Direct Applications

5.1 Theoretical Ouroboros

Closure

Ontological principle is proven as the inevitable consequence of geometric consistency. Field Equation $\Longleftrightarrow$ Ontological Principle

We have shown that this single Ontological Principle (2.13), through pure geometric reasoning, necessarily leads to an equation which mathematically expresses the very same equivalence we began with. We started with SPACETIME $\equiv$ ENERGY, from which geometry and physical laws are logically derived, and these derived laws then loop back to intrinsically define and limit the very nature of energy and spacetime, proving the self-consistency of the initial idea.

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY.}}$$

The ratio of geometric scales equals the ratio of energy densities.

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho}{\rho_{max}}}$$

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$$\boxed{\begin{array}{c}\text{SPACETIME GEOMETRY}\\ \equiv \\ \text{ENERGY DISTRIBUTION}\end{array}}$$

Summary

All physical structure emerges from the single relational equivalence:

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$$

From this, by enforcing geometric self-consistency, one necessarily arrives at the Unified Geometric Field Equation:

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho}{\rho_{\max}}.}$$

This is not an external law but an intrinsic closure relation: geometry and energy are two mutually defining projections of a single entity. It represents the completion of the theoretical Ouroboros — where the principle generates its own mathematical expression and the expression in turn validates the principle.

From a philosophical and epistemological point of view, this can be considered the crown achievement of any theoretical framework - the "Theoretical Ouroboros". But regardless of the aesthetic beauty of this result, let us remain sceptical.

5.2 $W_{\mathbf{ILL}}$: Unity of Relational Structure

The ontological principle

$$\text{SPACETIME} \equiv \text{ENERGY}$$

states that there is only one closed relational capacity - WILL. What we call space, time and energy are different projections of the same relational structure. For any energy-closed system observed from a relational origin, this resource appears through four operational projections:

Desmos project

WILL= 1 Derivation of the W_{ILL} invariant.

These quantities are defined as correlated projections of the same underlying WILL structure. In the dimensionful form we write:

$$M = \frac{\beta^2}{\beta_Y}\frac{c^2 a}{G}, \qquad E = \frac{\kappa^2}{\kappa_X}\frac{c^4 a}{2G}, \qquad T = \kappa_X(\frac{2Gm_0}{\kappa^2 c^3})^2, \qquad L = \beta_Y(\frac{Gm_0}{\beta^2 c^2})^2,$$

where (β, β_Y) and (κ, κ_X) are the kinematic and gravitational projections on the carriers S^1 and S^2, m_0 is the rest mass, $E_0 = m_0 c^2$ is the rest energy, and a is the relational scale of the system as semi-major axis (average length per cycle within this system).

The relational conservation of the carriers $\beta^2 + \beta_Y^2 = 1, \quad \kappa_X^2 + \kappa^2 = 1$, together with the closure theorem $\kappa^2 = 2\beta^2$, fix a unique dimensionless combination of these four projections. Combining E, T, M, and L we obtain

$$\text{W}_{\text{ILL}} \equiv \frac{ET}{ML} = \frac{\frac{E_0}{\kappa_X}\kappa_X t^2}{\frac{m_0}{\beta_Y}\beta_Y a^2} = \frac{E_0 t^2}{m_0 a^2}.$$

By the relations that tie temporal $t = a/c$ and spatial $a = R_s/\kappa^2$ scales this ratio is identically equal to unity for any relationally closed system:

$$\boxed{\text{W}_{\text{ILL}} = \frac{ET}{ML} = 1.}$$

All dimensionful constants cancel automatically; the value is fixed by the geometry of the carriers, not by a choice of units.

Desmos project

WILL EARTH GPS calculates GPS satellite time shift and testing $W_{ILL} = 1$ on the Earth-GPS system

The same invariant can be written in a phase–normalized form, using local projections

$$E_o = \frac{E_0}{\kappa_{Xo}}, \qquad M_o = \frac{m_0}{\beta_{Yo}}, \qquad T_o = \kappa_{Xo} t_o^2, \qquad L_o = \beta_{Yo} r_o^2,$$

Equivalently, for any state (β_o, κ_o), any scale r_o and any phase along the orbit. Then

$$\boxed{W_{\text{ILL}} = \frac{E_o T_o}{M_o L_o} = 1 \quad \text{for ALL ENERGY'S, ALL SCALES and ALL PHASES.}}$$

$$\frac{E_o}{M_o} = \frac{L_o}{T_o},$$

so the energy sector and the spacetime sector are not independent. Every change of the relational state rescales (E_o, M_o) and (T_o, L_o) coherently so that this equality is always preserved. The familiar practice of treating energy–mass and space–time as separate blocks is therefore an interpretational approximation. This equation does not bring any new predictive power but the ontological meaning shows that any coherent unit system will naturally collapse in to unity dissolving the false separation:

$$\text{SPACETIME} \equiv \text{ENERGY}$$

$$\boxed{W_{ILL} = 1}$$

One conserved relational resource appears as mass, energy, time and length, but always in a way that keeps their ratio fixed.

5.2.1 Interpretive Note: The Name "WILL"

The term **WILL** stands for **SPACE-TIME-ENERGY**. It is both a formal shorthand and a philosophical statement: the universe is not a stage where energy acts through time upon space, but a single self-balancing structure whose internal distinctions generate all phenomena. The name also serves as a gentle irony toward anthropic thinking: the Cosmos does not possess "will" - yet through WILL, it manifests All that Is.

Summary

$$\boxed{\textbf{WILL} \equiv \frac{ET}{ML} = 1 \quad \Longleftrightarrow \quad \text{Geometry} = \text{Energy} = \text{Causality.}}$$

WILL is not the unit of something - but the Unity of Everything.

5.3 From the 1PN Lagrangian to R.O.M.: A Falsifiable Closed-Form Alternative

Having shown in section "Two-Point Origin of the Lagrangian and Hamiltonian" (4.3) that the Lagrangian and Hamiltonian are artefacts of forcing $r_A = r_B$, we now demonstrate that the relational ontology supplies a complete, non-perturbative replacement. Starting from the standard 1PN Lagrangian only as a reference point, we translate it into the dimensionless projections β, κ and impose the closure (thm:closure) condition (cor:condition) that follows from methodological principles (sec:foundational) of Relational Geometry.

In standard celestial mechanics the relativistic two-body problem is treated by appending post-Newtonian (PN) corrections to the Newtonian Lagrangian or Hamiltonian. The equations of motion are then solved perturbatively, and observables such as the periastron advance emerge as power series in v/c and GM/rc^2. The complexity grows combinatorially with each PN order.

Relational Geometry replaces this entire perturbative machinery by a *finite* set of dimensionless algebraic relations that classify the allowed bound states of a two-body system. These relations are not a truncated series; they are exact consequences of the closure of the underlying relational carriers. Importantly, R.O.M. is *not* a re-summation of the GR PN expansion. It yields a closed-form periastron advance that agrees with GR at first post-Newtonian order but differs at second and higher orders — a deviation that is currently below observational thresholds and thus provides a sharp, falsifiable prediction.

This section gives an explicit, step-by-step mapping from the standard 1PN Lagrangian to the R.O.M. algebraic system and highlights exactly where the two methods coincide and where they diverge.

5.3.1 The standard 1PN test-mass Lagrangian

For a test particle of mass m moving in the field of a central mass M, the Lagrangian in harmonic coordinates, expanded to first post-Newtonian order and divided by m, is

$$L = -c^2 + \frac{1}{2}v^2 + \frac{GM}{r} + \frac{1}{8c^2}v^4 + \frac{3}{2c^2}\frac{GM}{r}v^2 - \frac{1}{2c^2}\left(\frac{GM}{r}\right)^2 + \mathcal{O}(c^{-4}). \tag{46}$$

The terms are, in order: rest energy, Newtonian kinetic and potential energy, the first relativistic correction to kinetic energy, the kinetic-potential coupling, and the non-linear correction to the potential. From this Lagrangian one derives equations of motion and integrates them perturbatively to obtain, for example, the periastron advance

$$\Delta\varphi_{1\text{PN}} = \frac{3\pi R_s}{a(1-e^2)}, \qquad R_s = \frac{2GM}{c^2}.$$

5.3.2 Translation into R.O.M. projections

R.O.M. operates with two dimensionless, frame-invariant projections that are directly measurable from spectroscopic and astrometric data sec:spectroscopic shift without invoking G or M.

$$\beta \equiv \frac{v}{c}, \tag{47}$$

$$\kappa^2 \equiv \frac{R_s}{r}. \tag{48}$$

In terms of these projections, the Newtonian potential term becomes $\dfrac{GM}{r} = \dfrac{1}{2}c^2\kappa^2$. Switching to units $c = 1$ for brevity, the 1PN Lagrangian rewrites as

$$L = \frac{1}{2}\beta^2 + \frac{1}{2}\kappa^2 + \frac{1}{8}\beta^4 + \frac{3}{4}\kappa^2\beta^2 - \frac{1}{8}\kappa^4 + \mathcal{O}(\beta^6). \tag{49}$$

This expression is still a perturbative expansion; the true trajectory must be obtained by varying the Lagrangian and solving differential equations.

5.3.3 The R.O.M. closure condition and why the Lagrangian is no longer required.

The foundational role of **closure theorem** (3.20): For any relationally closed the amplitudes β and κ are not independent but satisfy closure condition (3.23):

$$\boxed{\kappa^2 = 2\beta^2}. \tag{50}$$

This relation follows from the equal distribution of the total conserved relational resource among the independent degrees of freedom of the kinematic carrier S^1 (1 DOF) and the potential carrier S^2 (2 DOF). Closure is not an equation of motion; it is a global structural constraint that every admissible orbit must fulfil.

Once closure is imposed, the Lagrangian is no longer required. The permissible orbital states are completely determined by the algebraic consequences of (5.3.3) together with two further invariants that are already implicit in the Newtonian two-body problem (the conservation of the relational binding energy $\beta^2 = \kappa_o^2 - \beta_o^2$ and of the angular momentum $h = r_o\beta_T c$). The result is a closed system of finite, exact identities.

5.3.4 The R.O.M. algebraic system (exact within the relational framework)

For an elliptical orbit with eccentricity e and semi-major axis a, R.O.M. yields the following closed-form relations (true anomaly O):

- **Keplerian architecture (exact, not just Newtonian):**

$$\frac{R_s}{a} = 2\beta^2 \implies a = \frac{R_s}{2\beta^2}, \qquad T = \frac{2\pi a}{\beta c} = \frac{\pi R_s}{\beta^3 c}. \tag{51}$$

- **Orbital shape (the same ellipse, but with a precessing O):**

$$r(O) = \frac{a(1-e^2)}{1 + e\cos O}. \tag{52}$$

- **Periastron precession per orbit (exact R.O.M. expression):**

$$\Delta\varphi_{\rm ROM} \;=\; \frac{2\pi\,\tau_Y^2}{1-e^2} \;=\; \frac{2\pi\big(3\beta^2-2\beta^4\big)}{1-e^2}. \tag{53}$$

- **Time evolution (the same Kepler integral, with the period from (5.3.4)):**

$$t(O) \;=\; \frac{T}{2\pi}\,\zeta(O), \qquad \zeta(O) = \frac{1}{\sqrt{1-e^2}}\int_0^O \frac{d\theta}{(1+e\cos\theta)^2}. \tag{54}$$

These relations are algebraically exact within R.O.M.; they contain no free parameters and no perturbative approximations. The only departure from Newtonian mechanics is the precession formula (5.3.4).

5.3.5 Connection to the 1PN Lagrangian and the point of departure from GR

Substituting $\kappa^2 = R_s/a$ and the closure condition $\beta^2 = \kappa^2/2$ into (5.3.4) yields the equivalent form

$$\Delta\varphi_{\rm ROM} \;=\; \frac{3\pi R_s}{a(1-e^2)} \;-\; \frac{\pi R_s^2}{a^2(1-e^2)}. \tag{55}$$

The first term is exactly the GR 1PN advance (5.3.1). Thus, at first post-Newtonian order R.O.M. and GR coincide perfectly—the 1PN Lagrangian is therefore the leading-order content of the R.O.M. algebraic system.

The second term, $-\pi R_s^2/[a^2(1-e^2)]$, is the R.O.M. correction at $\mathcal{O}(\beta^4)$. It is *not* equal to the standard GR 2PN periastron advance; it constitutes a specific, built-in prediction of the relational framework. For Mercury this deviation is -7.3×10^{-7} arcseconds per century, roughly 10^{-8} of the total precession and far below current measurement precision. In stronger fields (e.g., the Galactic Centre S-stars, binary pulsars) the difference will become proportionally larger and will eventually be tested by accurate enough observations.

Falsifiability

R.O.M. does *not* claim to reproduce the full infinite PN series of GR. Instead, it offers a mathematically transparent alternative: a single, closed-form algebraic expression for the periastron advance that agrees with GR at 1PN but makes a definite, non-adjustable prediction at 2PN and beyond. Future high-precision measurements of periastron precession (e.g., from GRAVITY+ observations of S2, or from binary pulsar timing) will either confirm this prediction or rule out the R.O.M. closure law.

5.3.6 Summary of the mapping

1PN Lagrangian (harmonic coordinates)	**R.O.M. algebraic system**
$L = -c^2 + \frac{1}{2}v^2 + \frac{GM}{r} + \frac{v^4}{8c^2} + \frac{3}{2c^2}\frac{GM}{r}v^2 - \frac{1}{2c^2}\left(\frac{GM}{r}\right)^2 + \dots$	$\kappa^2 = \frac{R_s}{r}, \quad \beta = \frac{v}{c}$ Closure: $\kappa^2 = 2\beta^2$ Orbit: $r(O) = \frac{a(1-e^2)}{1+e\cos O}, \quad a = \frac{R_s}{2\beta^2}$ Precession: $\Delta\varphi_{\rm ROM} = \frac{2\pi(3\beta^2-2\beta^4)}{1-e^2}$ Time: $t(O) = \frac{T}{2\pi}\zeta(O), \quad T = \frac{\pi R_s}{\beta^3 c}$
Connection: The GR 1PN advance is the leading term of $\Delta\varphi_{\rm ROM}$; the R.O.M. expression adds a specific $\mathcal{O}(\beta^4)$ correction that differs from GR 2PN and provides a falsifiable signature of the relational framework.	

Key takeaway

The 1PN Lagrangian is a truncated series in the small parameters β^2 and κ^2. R.O.M. replaces it with a finite algebraic system that is exact within the relational ontology, contains no free parameters, and matches GR at leading order while making a definite, testable prediction at the next order. The mapping is mathematically transparent and eliminates the need for perturbative expansions altogether.

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5.4 Methodology comparison of General Relativity (GR) with WILL Relational Geometry (R.O.M.)

Webpage Link

https://willrg.com/RGvsGR/

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5.5 Discussion

All WILL RG results, including the success of R.O.M., remain a single unbroken generative chain of derivations from methodological constraints with zero flexibility. Relational Orbital Mechanics functions as the direct operational consequence of applying the Foundational Methodological Principles to bound gravitational systems.

5.5.1 Epistemic Benefits

WILL RG enforces epistemic hygiene by operating exclusively on dimensionless, operationally defined observables. By replacing the absolute background manifold with the topological closure of the S^1 and S^2 carriers, the framework eliminates surplus ontology — absolute space, absolute time, and independent mass. Every parameter in the algebraic chain (β, κ, e, T, z) maps directly to a measurable spectroscopic or chronometric quantity. Consequently, the theory achieves maximum ontological parsimony: it posits only what is structurally necessary to close the relational ledger, stripping away all metaphysical artifacts inherited from classical continuum mechanics.

Operational Benefits: Absence of Differential Formalism and Acceleration

The most immediate operational consequence of the relational construction is the absence of differential equations of motion and the concept of acceleration. In standard orbital mechanics the central acceleration (Newtonian or geodesic) is the primitive from which orbital period, shape, precession, and equilibrium configurations are derived. In WILL RG the same observables are obtained directly from algebraic identities among the projections on the carriers S^1 and S^2, once the methodological principles and the unifying principle SPACETIME $\equiv$ ENERGY have fixed the geometry of those carriers and the unique exchange rate between them.

This absence operates as a necessary consequence of the generative structure. Because the theory begins by removing the separation between structure and dynamics, and because it enforces topological closure and maximal symmetry from the outset, the allowed relational states of a bound system are completely classified by the dimensionless projections and the single closure condition that the methodology itself produces. The mathematics requires no integration of trajectories, no effective potentials, and no auxiliary constructs such as rotating frames. It remains a short sequence of explicit algebraic relations whose every step retains clear physical meaning.

The gain in transparency is therefore structural. Observables that in conventional treatments appear only after lengthy perturbative expansions or numerical integration of differential equations emerge here as immediate, closed-form consequences of the relational geometry. This is visible across all derivations in the paper, including the period–semi-major-axis relation, eccentricity, secular precession, light deflection, and the location of the Sun–Earth L1 point (Sections Relational Parameterization and Sun–Earth L1 point).

5.5.2 Ontological Shift: From Descriptive to Generative Physics

In conventional physics the methodology follows a descriptive paradigm:

1. Observable phenomena are identified.
2. Empirical regularities are codified as "laws of nature."
3. Mathematical formalisms are constructed to *describe* these regularities.

Thus, physical laws enter as external assumptions that model what is seen. Even in General Relativity, where geometry plays the central role, the equivalence principle and the metric postulate remain external inputs.

RG inverts this paradigm. Laws are generated as inevitable consequences of relational geometry:

- Concepts such as "inertial mass equals gravitational mass" emerge automatically as algebraic identities enforced by the geometry.
- What classical physics calls "laws of nature" serve as secondary shadows of the single relational principle:

$$\text{SPACETIME} \equiv \text{ENERGY}.$$

Summary

Standard Physics: Laws *describe* what we observe.
Relational Geometry: Laws are *generated* as necessary products of closure and self-consistency.

In this sense, the ontological role of physical law is transformed. Physics ceases to be a catalog of empirical descriptions, and becomes the logical unfolding of a single relational structure. WILL identifies the necessary conditions under which all observed phenomena arise.

Descriptive Physics (Standard)	**Generative Physics (WILL)**
Phenomena are *observed* first, then summarized into empirical laws.	Laws emerge as *inevitable consequences* of relational geometry.
Physical laws are *assumptions* introduced to model reality.	Physical laws are *identities*, enforced by geometric self-consistency.
Time and space are treated as external backgrounds.	Time and space are *projections of energy relations*.
Dynamics = evolution of states *in time*.	Dynamics = ordered succession of balanced configurations; *time is emergent*.
Goal: *describe* what is observed.	Goal: *show why nothing else is possible*.

Table 5: Ontological contrast between standard descriptive physics and the generative paradigm of WILL Relational Geometry.

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Phenomenon	Standard (GR) Result	Relational Geometry (RG)
GPS time shift / gravitational redshift	Frequency shift = combination of kinetic (SR) and gravitational (GR) effects.	Single symmetric law: $\tau = \beta_Y \cdot \kappa_X$, $E_{\text{loc}} = \frac{E_0}{\sqrt{(1-\beta^2)(1-\kappa^2)}} = \frac{E_{\text{loc}}}{\tau}$ 38.52 μs/day verified directly with GPS satellites.
Photon sphere, ISCO, horizons	Derived by solving geodesic equations in Schwarzschild metric.	Critical radii emerge from simple symmetries (Photon sphere: $\theta_1 = \theta_2 = 54.73°$ ("magic angle") $Q^2 = \kappa^2 + \beta^2 = 1$. ISCO: $Q^2 = Q_Y^2 = 1/2$).
Mercury's perihelion precession	Complex expansion of Einstein field equations.	Same number obtained from RG with $\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1-e^2} = 43''$/century. Using simple algebra.
Cosmological redshift	Photon "loses energy" as universe expands.	Energy conserved; redshift = redistribution of projection parameters. (Details in WILL PART II)
Cosmological absolute scale (Supernovae fit)	Hubble-like expansion, ΛCDM fits	$H_0 \equiv \sqrt{8\pi G \rho_\gamma/(3\alpha^2)} = 68.15$ Derived from CMB temperature and α connecting micro and macro scales (Details in WILL PART II-III).
Cosmological constant Λ	Added by hand to fit data ("dark energy").	Arises naturally as $\Lambda = 2/3r^2$. No extra entities required. (More details in WILL PART I and II)
Singularities	Predicted in black holes and big bang ($\rho \to \infty$).	Structurally absent: density bounded by $\rho_{\max} = c^2/(8\pi G r^2)$.
Local gravitational energy	Indirectly localized at infinity (ADM/Bondi).	Directly measurable via κ, e.g. from light deflection angle or red shift.
Unification with QM	Absent in standard GR framework.	Same projectional law applies from microscopic $\alpha = \beta_1$ (QM) to cosmic $\kappa^2 = \Omega_\Lambda$ (GR, COSMO) scales. (Details in WILL PART II and III)

Table 6: Classical GR results vs. WILL RG outcomes. Known effects are recovered by simpler symmetric laws, while new predictions eliminate singularities and explain cosmology without dark entities.

5.6 Challenges and Current Limitations

Gravitational waves and the full radiative two-body problem remain to be formulated within the relational framework. The current construction addresses only conservative, bound orbits. Extending it to dissipative, radiative dynamics requires a relational description of propagating degrees of freedom on the carriers and remains an open technical task.

A complete, self-consistent multi-body architecture that extends beyond the collinear L1 case awaits explicit algebraic realisation. The nested-contour approach provides a clear conceptual route, but its explicit mathematical formulation for systems containing three or more gravitationally bound bodies (including mean-motion resonances and long-term stability) requires further development. The author states openly that progress on both fronts has so far been limited by present mathematical facility rather than by any intrinsic prohibition within the relational ontology. Collaboration with researchers experienced in radiative dynamics and in the algebraic treatment of multi-body closures is therefore actively invited.

5.7 Falsifiability and Future Tests

Relational Orbital Mechanics makes several sharp, parameter-free predictions that can be tested with improved observational precision.

The most direct test comes from continued high-precision monitoring of the S2 star. Current GRAVITY data are statistically consistent with both RG and GR. Future pericentre passages will deliver significantly higher astrometric precision. These observations will probe the detailed phasing of the apsidal advance and the magnitude of higher-order corrections, allowing a direct comparison between the exact algebraic predictions of RG and the corresponding GR results. Specifically the difference in second order precession term $\mathcal{O}(\beta^4)$.

The higher-order correction is fixed directly by the closure relation. It contain no free parameters. A detection at the predicted level, or a sufficiently tight upper limit below it, would therefore test the relational model itself.

5.8 Conclusion

Relational Geometry demonstrates that the entire phenomenology of motion – from Keplerian architecture to perihelion precession and light deflection – emerges as an inevitable algebraic consequence of the Foundational Methodological Principles. All WILL RG results, including the complete success of R.O.M., remain an unbroken chain of derivations from first principles under extreme methodological constraints with zero flexibility.

This operates as the irreducible algebraic core that remains when the ontological superstructure of Riemannian geometry and tensorial formalism is removed. The S2 orbit fit and Mercury's precession clearly show that RG functions as a distinct model with radically cleaner and lighter ontological baggage. Future, more accurate astronomical data will definitively show which ontology the empirical data prefers.

Within RG several structural consequences follow:

1. **G and M are secondary.** The Schwarzschild length R_s and all orbital distances are derived directly from dimensionless spectroscopic and chronometric ratios. The Newtonian gravitational constant and the kilogram mass serve strictly as conversion factors when translating geometric results into legacy units.

2. **Derivation of Metrics from Relational Geometry.** The Minkowski, Schwarzschild, and Kerr metrics and their intervals are revealed as the Pythagorean identities of the relational carriers S^1 and S^2. Relativistic effects—such as time dilation and length contraction—emerge naturally as geometric rotations between external interaction (Amplitude) and internal existence (Phase).

3. **Closure replaces the virial theorem.** The relation $\kappa^2 = 2\beta^2$ emerges as the inevitable consequence of equitable capacity distribution across the independent degrees of freedom of the kinematic and potential carriers.

4. **Precession is topological, rather than dynamical.** The secular advance of the pericentre arises from the mismatch of internal phase over one cyclic extension of the orbital phase, bypassing the integration of geodesic equations or expansion in powers of v/c.

5. **Singularities are avoided** by naturally constraining the valid range of radial distance. **The point $r = 0$ remains absent from the admissible relational range.**

6. WILL trilogy offers a unified, parameter-free ground-up unbroken chain derivation of most major observed phenomena within extreme methodological constraints and zero flexibility. The traditional unjustified separation of spacetime and energy dissolves into a single generative structure we call WILL.

The results presented here invite exacting numerical tests (additional binary pulsars, triple systems, high-resolution S2 pericentre passages) and provide an operational language in which gravitational phenomena can be communicated without surplus ontology. The hope is that this physical and mathematical clarity returned by RG's methodology will help turn the focus of physical science back to its earliest ambition: to understand the Universe, rather than just to model it.

Final Summary

$$\textbf{SPACETIME} \equiv \textbf{ENERGY}.$$

References

[1] Schwarzschild, K. (1916). *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften.*

[2] Tolman, R.C. (1939). *Phys. Rev.* **55**, 364; Oppenheimer, J.R., Volkoff, G.M. (1939). *Phys. Rev.* **55**, 374.

[3] Dyson, F.W., Eddington, A.S., Davidson, C. (1920). *Phil. Trans. R. Soc. A* **220**, 291-333.

[4] Lebach, D.E. et al. (1995). *Phys. Rev. Lett.* **75**, 1439.

[5] Ashby, N. (2003). *Living Rev. Relativity* **6**:1.

[6] Will, C.M. (2014). *Living Rev. Relativity* **17**:4.

[7] Williams, J.G., Turyshev, S.G., Boggs, D.H. (2006). *Adv. Space Res.* **37**, 67-71.

[8] Kramer, M. et al. (2021). *Phys. Rev. X* **11**, 041050.

[9] Misner, Thorne, Wheeler (1973). *Gravitation*.

[10] *et al* 2022 *ApJ* 933 49 DOI 10.3847/1538-4357/ac752f

[11] Espenak, F. (2014). Seven Century Catalog of Mercury Transits: 2001 to 2700. *NASA Eclipse Web Site*.

[12] Williams, M. (2016). How Far is Mercury From the Sun? *Universe Today* (citing NASA data).

[13] Mamajek, E.E. et al. (2015). IAU 2015 Resolution B3 on Recommended Nominal Conversion Constants. *arXiv:1510.07674*.

[14] Meftah, M., et al. (2014). *Astron. Astrophys.* **569**, A15.

[15] Williams, D.R. (2024). *NASA Planetary Fact Sheet*. Goddard Space Flight Center. https://nasa.gov

[16] Folkner, W.M., et al. (2014). *IPN Progress Report* **42-196**, 1-81.

[17] Gonzalez Hernandez, J.I., et al. (2020). *Astron. Astrophys.* **643**, A146.

[18] NASA Radio JOVE Project, *Jupiter Fact Sheet*.https://radiojove.gsfc.nasa.gov/education/jupiter/basics/jfacts.htm, 13 July 2000.